

PROPOSED RESIDENTIAL BUILDING
FOR

Manita Kafle

Address: Lalitpur -7

STRUCTURAL ANALYSIS
&
DESIGN Report

Prepared By:

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PULCHOWK, LALITPUR

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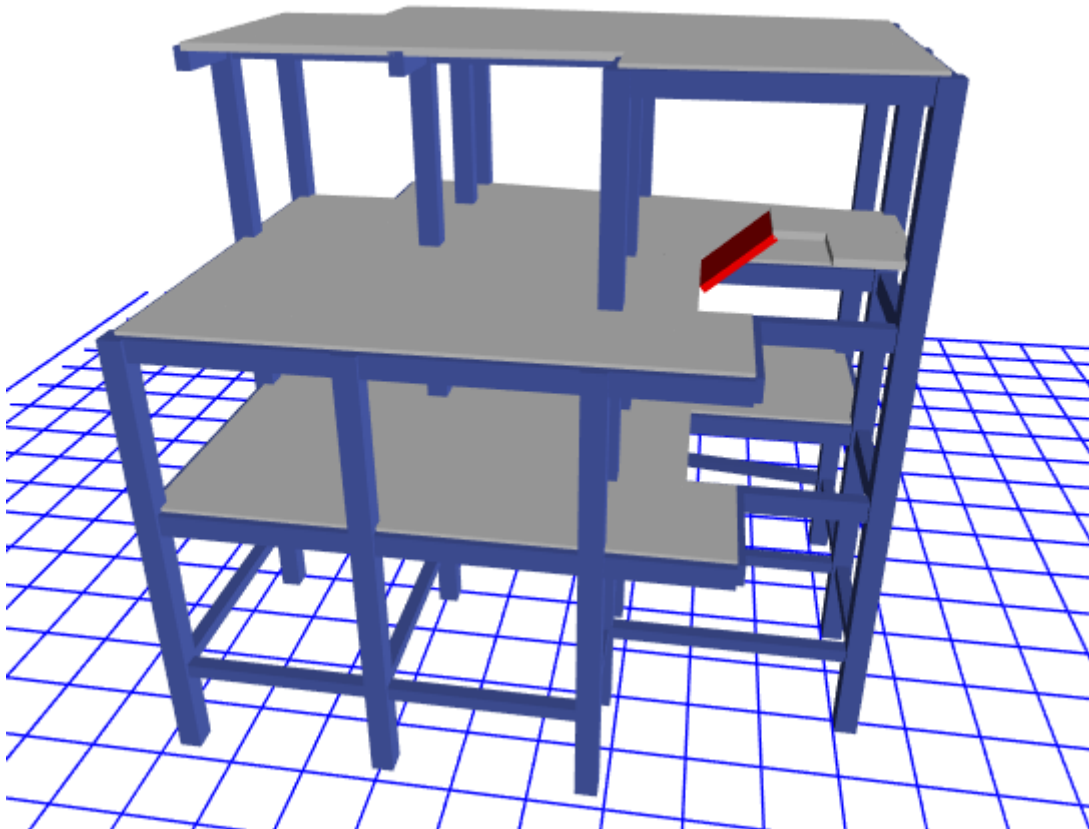


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1. Introduction

This report deals with the methodology of the Structural Analysis and Design of the Residential building at Lalitpur-7.

Project Information

Client	:	Manita Kafle
Building Type	:	Residential Building
Location	:	Lalitpur-7
Plinth Area	:	80.3611 sq.m

Building Features

Type of Structure	:	RCC Framed Structure
Frame System	:	Special Moment Resisting Frame
Floor System	:	Two-way Solid Slab
Storey	:	3
Extension Provision	:	None
Storey Height	:	1.6764m below Ground, 2.8448m for Ground & 1 st , 3.4036m for 2 nd Floor.
Total Height	:	10.7696m from the top of Foundation

Site Condition

Soil Type	:	D (for seismic consideration as per NBC 205:2077)
Seismic Zone	:	Kathmandu
Soil bearing capacity	:	130 KN/m ² (assumed)

2. Basic Data

a. Density of Concrete	=	25 kN/m ³
b. Live Load	=	2kN/m ² for Accessible Floor
	=	3kN/m ² for Storage room
	=	3kN/m ² for corridor
	=	3kN/m ² for Staircase
	=	1.5KN/m ² for Terrace
c. Floor Finishing Load	=	1kN/m ²
d. Density of Brick wall	=	18.85kN/m ³
e. Soil Bearing Capacity (adopted)	=	130 kN/m ²

3. Relevant Code followed

- NBC 105:2077 : Seismic Design of Buildings of Nepal
- IS 456-2000 : For Design of Structural Elements
- SP 16 : For Design Aid
- IS 13920:2016 : Ductile Detailing
- NBC 1994: National Building Code of Nepal

4. Software used for Analysis and Design

- ETABS 2018

5. Concrete and Steel Grade

Concrete Grade = M 20 for Footings, Beams, slabs & Columns

Steel Grade = Fe 500

6. Preliminary Design for proportioning of the Structural Elements

The tentative sizes of the Structural elements are determined through preliminary design so that after analysis, the presumed dimensions may not deviate considerably, thus making the final design both safe and economical. The tentative sizes have been determined as follows:

Slab:

From Deflection Criteria [Effective Depth = Effective Shorter Span / (26 * MF)]

Beam:

For Practical rule as 25mm Depth for 300mm of Span covering deflection Criteria

Column:

From evaluation of approximate gravity loading coming up to the critical Column.

To compensate for the possible eccentric Loading and earthquake loads the size is increased by about 50 % in design.

7. Loads on Structures

The following Loads were assumed to occur in Structural System.

- a. Dead Load
- b. Live Load
- c. Seismic Load

Dead Load:

Dead Load on the structure comprise the self-weight of the member; weight of the finishes and partition walls. These are usually dependent upon the constructional features and have to be assumed in order to design various structural concrete members. The Loads on the beams due to the slabs were calculated according to clause 23.5 of IS 456 – 2000. The Wall Load is taken for thickness of either 230 mm or 115 mm as per Architectural Drawing and suitable reduction is made for Window and Door Opening.

As the software, we have used, generates the self weight of the Structural member by itself, we have not calculated the self weight.

Live Loads or Imposed Loads:

The Live Load is taken as 2kN/m^2 for accessible floor and 3kN/m^2 for staircase.

Seismic Load:

For Earthquake Load, 100% of Dead Load and 30% of Live Load and 0% of Roof Live Load is taken into account.

Seismic Coefficient method using Nepal Building Code: NBC 105:2077 is applied for Earthquake Analysis of the Structure and the Parameters taken are:

Building Elements	As Per Submitted Design
Importance Factor (I)	1
Seismic Zone	Kathmandu
Zone Factor (Z)	0.35
Height of the building (h)	10.7696m
Time Period (T_x) (in seconds) $= 1.25 * 0.075h^{0.75}$	0.557341
Time Period (T_y) (in seconds) $= 1.25 * 0.075h^{0.75}$	0.557341
Soil Type	D (For Kathmandu Valley, very soft Soil)
Ductility Factor (R_μ)	4 (RCC Frame)
Maximum Total Deflection	31.13mm
Maximum Inter-Storey Drift Ratio	0.004289

Effective stiffness of cracked sections:

A rational analysis shall be performed in arriving at the elastic flexural and shear stiffness properties of cracked concrete and masonry elements. In absence of such analysis, the effective stiffness of cracked sections shall be taken from Table below:

S No.	Component	Flexural Stiffness	Shear Stiffness
1	Beam	$0.35 E_c I_g$	$0.40 E_c A_w$
2	Columns	$0.70 E_c I_g$	$0.40 E_c A_w$
3	Wall—cracked	$0.50 E_c I_g$	$0.40 E_c A_w$
4	Wall—uncracked	$0.80 E_c I_g$	$0.40 E_c A_w$

For steel structures, the gross stiffness values shall be used.

Load Combinations

Following Load Combinations are considered for critical conditions as per NBC 105:2077

1. $1.5DL + 1.5LL$
 2. $1.2DL + 1.5LL$
 3. $DL + \lambda LL \pm EQ$ Ultimate
 4. $DL + \lambda LL \pm EQ$ Service
- Where, $\lambda =$ 0.6 for storage Building
0.3 for Other Usage

8. Design Methods of Structural Elements

We have followed Indian Standard Code of Practice for Plain and Reinforced Concrete IS: 456-2000 for design of Structural Elements. This incorporates the two methods of Structural Design of RC structures specified as:

- Working Stress Method* based on the Working loads in conjunction with permissible stresses in the materials.
- Limit State Method* based on safety and serviceability requirements associated with the design loads and design strengths of the materials. These design loads and design strengths are obtained by applying partial safety factors for characteristic loads and strengths of the materials concrete and steel.

We have followed the limit state method which is incorporated in IS:456-2000. It is consistent with the new philosophy of design termed *limit state approach* which was incorporated in the Russian Code – 1954, the British code BS 8110 – 1985 and the American Code ACI 318 – 1989.

Limit State Method

- **Limit States**

The Limit State method of design covers the various forms of failure. There are several limit state at which the structure ceases to function, the most important among them being,

- The limit state of collapse or total failure of structure.
It corresponds to the maximum load carrying capacity. Violation of collapse implies failure. This limit state corresponds to Flexure, Compression, Shear and Torsion.
- The limit state of serviceability which includes excessive deflection and excessive local damage.
Excessive deflection adversely affects the finishes and excessive local damage results in cracking of concrete, which impairs the efficiency, or appearance of the structure.

Design Equations in Limit State Method :

Singly Reinforced Sections:

The depth of Neutral Axis x_u is obtained as

$$X_u = \left[\frac{0.87 A_{st} f_y}{0.36 f_{ck} b} \right]$$

For under Reinforced Sections, the neutral Axis depth X_u is limited to a value of

$$X_{u, \max} < 0.48 d \text{ for steel of } f_y = 500 \text{ N/mm}^2$$

If the neutral axis depth is less than these limiting values, the ultimate moment is computed by the equation

$$M_u = 0.87 f_y A_{st} d \left[1 - \left\{ \frac{f_y A_{st}}{b d f_{ck}} \right\} \right]$$

If the Neutral Depth is equal to the limiting values, then the section is balanced or limiting and the moment capacity is evaluated by the equation.

$$M_{u, \lim} = 0.36 X_{u, \max} / d \left[1 - 0.42 X_{u, \max} / d \right] b d^2 f_{ck}$$

$$M_{u, \lim} = 0.138 f_{ck} b d^2 \text{ (for HYSD bars with } f_y = 500 \text{ N/mm}^2 \text{)}$$

Area of steel in a balanced section is given by:

$$A_{st} = \left[\frac{0.36 f_{ck} b X_{u, \max}}{0.87 f_y} \right]$$

Doubly Reinforced Sections:

The ultimate moment resistance of sections with compression reinforcement (doubly reinforced Section) is computed by the equation.

$$[M_u - M_{u,lim}] = f_{sc} A_{sc} (d - d')$$

Where, M_u = Moment of Resistance of the doubly reinforced sections
 $M_{u,lim}$ = Limiting moment of Resistance of a section without Compression reinforcement
 d = Effective Depth
 d' = Depth of Compression Reinforcement from compression face
 f_{sc} = Design stress in compression reinforcement

The total Area of Tension Reinforcement is obtained as

$$A_{st} = A_{st1} + A_{st2}$$

Where,

A_{st1} = Total Area of Tensile Reinforcement
 A_{st2} = Area of Tensile Reinforcement for a singly Reinforced Section for $M_{u,lim}$
 $A_{sc} = A_{st} f_{sc} / 0.87 f_y$

c. The nominal shear stress T_v in beams of uniform depth is obtained as

$$T_v = V_u / b d$$

Where, V_u = Ultimate shear Force due to design loads

b = Breadth of member

d = effective depth

The permissible shear stress T_c in beam depends upon the percentage of longitudinal reinforcement. Where $T_c < T_v$, minimum shear Reinforcement is given by

$$A_{sv} / b S_v > 0.4 / f_y$$

If T_v exceeds T_c , but is less than $T_{c \max}$, then Shear Reinforcements are designed to take up the balanced shear $V_u - T_c b d$

The Strength of Shear Reinforcement V_{us} is calculated as

$$V_{us} = 0.87 f_y A_{sv} d / S_v$$

9. Analysis and Design of the Structural Elements:

The analysis has been carried out using a standard software package Etab 2018. The software is capable of carrying out a Three-Dimensional analysis. It is windows-based software. It has a user-friendly graphical input and output interface. It can account for the rigid zones at the beam - column junction. It has a capability to create a special joint at any grid intersection. The program has a facility to create shell elements also. Shell elements can be used for the analysis of Slabs and Shear walls.

A Three-Dimensional Linear elastic analysis has been carried out. A model based on Rigid Diaphragm Concept has been considered. This is done by creating a special joint at the center of mass of each floor level and constraining all the joints at this level by a diaphragm constraint. The stiffness contribution of brick walls on the structure has been ignored and the building has been modeled as a "Bare Frame" with no infill wall panels.

10. Column-Beam Moment Capacity Ratio

At every beam column junction in a frame, the summation of the moment capacities of the column end sections shall be greater than 1.2 times the summation of the beam end moment capacities.

$$\frac{\sum M_c}{\sum M_b} > 1.2$$

Where, $\sum M_c$ = sum of the design moment of resistance of the column above and below the beam centerline;

$\sum M_b$ = sum of the design moment of resistance of beams at the intersection of the beam and column centerline

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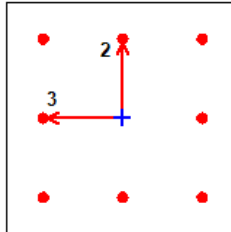


11. Design of structural members

DESIGNS

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Column Section Design



Column Element Details Type: Ductile Frame (Summary)

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF
First	C10	43	COLUMN (14 *14)	DL+0.3LL- EQyU	2489.2	2844.8	1

Section Properties

b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)
355.6	355.6	56	28

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
22360.68	20	1	500	500

Design Code Parameters

γ_c	γ_s
1.5	1.15

Axial Force and Biaxial Moment Design For P_u , M_{u2} , M_{u3}

Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M_2 kN-m	Minimum M_3 kN-m	Rebar Area mm ²	Rebar % %
56.7889	34.6246	13.6158	1.1358	1.1358	1012	0.8

Axial Force and Biaxial Moment Factors

	K Factor Unitless	Length mm	Initial Moment kN-m	Additional Moment kN-m	Minimum Moment kN-m
Major Bend(M3)	0.957569	2489.2	5.4463	0	1.1358
Minor Bend(M2)	0.931191	2489.2	13.8499	0	1.1358

Shear Design for V_{u2} , V_{u3}

	Shear V_u kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm ² /m
Major, V_{u2}	31.1384	53.2059	42.6154	31.1384	394.16
Minor, V_{u3}	26.3677	53.2059	42.6154	25.3709	394.16

Joint Shear Check/Design

	Joint Shear Force kN	Shear V_{Top} kN	Shear $V_{u,Tot}$ kN	Shear V_c kN	Joint Area cm ²	Shear Ratio Unitless
Major Shear, V_{u2}	0	22.2417	193.3185	565.5077	1264.5	0.342
Minor Shear, V_{u3}	0	18.1221	153.1129	565.5077	1264.5	0.271

(1.4) Beam/Column Capacity Ratio

Major Ratio	Minor Ratio
0.699	0.57

Additional Moment Reduction Factor k (IS 39.7.1.1)

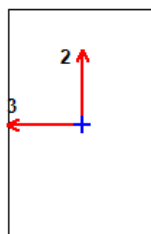
A_g cm ²	A_{sc} cm ²	P_{uz} kN	P_b kN	P_u kN	k Unitless
1264.5	10.1	1517.4163	457.7286	56.7889	1

Additional Moment (IS 39.7.1)

	Consider M_a	Length Factor	Section Depth (mm)	KL/Depth Ratio	KL/Depth Limit	KL/Depth Exceeded	M_a Moment (kN-m)
Major Bending (M_3)	Yes	0.875	355.6	6.703	12	No	0
Minor Bending (M_2)	Yes	0.875	355.6	6.518	12	No	0

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Beam Section Design



Beam Element Details Type: Ductile Frame (Summary)

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF
First	B17	64	BEAM (9 * 14)	DL+0.3LL+EQ xU	3124.2	3302	1

Section Properties

b (mm)	h (mm)	b _f (mm)	d _s (mm)	d _{ct} (mm)	d _{cb} (mm)
228.6	355.6	228.6	0	25.4	25.4

Material Properties

E _c (MPa)	f _{ck} (MPa)	Lt.Wt Factor (Unitless)	f _y (MPa)	f _{ys} (MPa)
22360.68	20	1	500	500

Design Code Parameters

γ _c	γ _s
1.5	1.15

Factored Forces and Moments

Factored M _{u3} kN-m	Factored T _u kN-m	Factored V _{u2} kN	Factored P _u kN
-40.523	4.2171	42.3497	0.6586

Design Moments, M_{u3} & M_t

Factored Moment kN-m	Factored M _t kN-m	Positive Moment kN-m	Negative Moment kN-m
-40.523	6.3394	0	-46.8624

Design Moment and Flexural Reinforcement for Moment, M_{u3} & T_u

	Design -Moment kN-m	Design +Moment kN-m	-Moment Rebar mm ²	+Moment Rebar mm ²	Minimum Rebar mm ²	Required Rebar mm ²
Top (+2 Axis)	-46.8624		354	0	354	162
Bottom (-2 Axis)		0	177	0	0	177

Shear Force and Reinforcement for Shear, V_{u2} & T_u

Shear V_e kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm ² /m
53.5935	0	96.4823	36.5249	809.69

Torsion Force and Torsion Reinforcement for Torsion, T_u & V_{U2}

T_u kN-m	V_u kN	Core b_1 mm	Core d_1 mm	Rebar A_{svt} /s mm ² /m
6.1277	38.2836	197.8	324.8	394.96

Column Design Summary

Column	Column Type	Column Size b X D	Column Naming	Column Area mm ²	Ast Required as per Etab DESIGN mm ²	Dia				Area Provided	% of Steel provided
						25	20	16	12		
BELOW GROUND FLOOR COLUMN : The area of steel calculated is according to Etab 2018											
B-1	Square	350*350	C1	126451.36	1329.00			8		1608.00	1.27
C-1	Square	350*350	C1	126451.36	1102.00			8		1608.00	1.27
D-1	Square	350*350	C1	126451.36	1131.00			8		1608.00	1.27
A-2	Square	350*350	C1	126451.36	1437.00			8		1608.00	1.27
B-2	Square	350*350	C1	126451.36	1043.00			8		1608.00	1.27
A-3	Square	350*350	C1	126451.36	1134.00			8		1608.00	1.27
B-3	Square	350*350	C1	126451.36	1012.00			8		1608.00	1.27
C-3	Square	350*350	C1	126451.36	1038.00			8		1608.00	1.27
D-3	Square	350*350	C1	126451.36	1110.00			8		1608.00	1.27
C-4	Square	350*350	C1	126451.36	1122.00			8		1608.00	1.27
D-4	Square	350*350	C1	126451.36	1440.00			8		1608.00	1.27
A-5	Square	350*350	C1	126451.36	1389.00			8		1608.00	1.27
B-6	Square	350*350	C1	126451.36	1150.00			8		1608.00	1.27
C-7	Square	350*350	C1	126451.36	1499.00			8		1608.00	1.27
GROUND FLOOR COLUMN											
B-1	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-1	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-1	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-2	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-2	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-4	Square	350*350	C1	126451.36	1049.00			4	4	1256.00	0.99
D-4	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-5	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-6	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-7	Square	350*350	C1	126451.36	1062.00			4	4	1256.00	0.99
FIRST FLOOR											
B-1	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-1	Square	350*350	C1	126451.36	1017.00			4	4	1256.00	0.99
D-1	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-2	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-2	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-4	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-4	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-5	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-6	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-7	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99

Column Design Summary

Column	Column Type	Column Size b X D	Column Naming	Column Area mm ²	Ast Required as per Etab DESIGN mm ²	Dia				Area Provided	% of Steel provided
						25	20	16	12		
SECOND FLOOR											
A-2	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-2	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-3	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
C-4	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-4	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
A-5	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
B-6	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99
D-7	Square	350*350	C1	126451.36	1012.00			4	4	1256.00	0.99

TABLE: Joint Reactions										
Grid	Label	Unique Name	Output Case	Case Type	FX	FY	FZ	MX	MY	MZ
					kN	kN	kN	kN-m	kN-m	kN-m
C-7	1	2	1.5DL+1.5LL	Combination	-9.3858	1.4946	274.59	-0.901	-4.685	-0.036
B-6	2	4	1.5DL+1.5LL	Combination	-4.6508	3.4713	535.43	-2.024	-2.321	-0.036
A-5	3	6	1.5DL+1.5LL	Combination	12.1777	4.2107	350.041	-2.57	6.0415	-0.036
C-4	4	8	1.5DL+1.5LL	Combination	5.5633	-0.825	357.921	0.2483	2.7846	-0.036
D-4	5	10	1.5DL+1.5LL	Combination	-4.7616	4.9819	267.718	-2.487	-2.332	-0.036
A-3	6	12	1.5DL+1.5LL	Combination	11.3025	-1.525	456.845	0.2722	5.7396	-0.036
B-3	7	14	1.5DL+1.5LL	Combination	-12.64	1.4654	690.672	-1.029	-6.125	-0.036
C-3	8	16	1.5DL+1.5LL	Combination	3.9216	5.8168	529.638	-3.043	2.082	-0.036
D-3	9	18	1.5DL+1.5LL	Combination	-2.4281	8.0972	451.756	-4.03	-1.065	-0.036
A-2	10	20	1.5DL+1.5LL	Combination	7.3591	-5.288	303.287	2.137	3.8887	-0.036
B-2	11	22	1.5DL+1.5LL	Combination	-8.2879	-2.727	485.666	1.0479	-3.865	-0.036
B-1	12	24	1.5DL+1.5LL	Combination	7.5752	-2.04	244.168	0.7078	4.0803	-0.036
C-1	13	26	1.5DL+1.5LL	Combination	-0.8452	-4.747	324.936	2.1922	-0.093	-0.036
D-1	14	28	1.5DL+1.5LL	Combination	-4.9	-12.39	252.627	6.1209	-2.102	-0.036

Structure Analysis & Design Report on Residential Building

Bearing Capacity of Soil= 130 KN/m²

Strength of steel (fy)= 500 N/mm²

Strength of concrete(fck)= 20 N/mm²

Size of column

L1=	0.35 m
B1=	0.35 m

ISOLATED FOOTING DESIGN

Footing Grid	Axial force Pu	Axial non factored load	Factor load Pu'	Actual Ast m2	Size of footing (m)	Adopted size				Total area	Net pressure acting upward	B.M at face	Effective depth (d)	Adopted depth	Area of tension steel Ast	Bar (dia) mm	Adopted spacing mm	Check for A _s	%age of steel	One way shear			Two way shear				Check for Shear
						L(m)	B(m)	L(ft)	B(ft)											Shear force V _u	Nominal stress T _v	T _c	V _u	b _o	T _v	T _c	
B-1	244.168	162.779	179.06	1.38	1.1747	1.52	1.52	5	5	2.31	158.52	41.23	99.1	550.0	173.2	12	125	OK	0.02	8.43	0.01	0.12	237.84	3600	0.12	1.12	OK
C-1	324.936	216.624	238.29	1.83	1.3528	1.52	1.52	5	5	2.31	210.96	54.87	114.4	550.0	230.9	12	125	OK	0.03	11.22	0.01	0.14	316.52	3600	0.16	1.12	OK
D-1	252.627	168.418	185.26	1.43	1.1958	1.52	1.52	5	5	2.31	164.02	42.66	100.8	550.0	179.3	12	125	OK	0.02	8.73	0.01	0.12	246.1	3600	0.12	1.12	OK
A-2	303.287	202.191	222.41	1.71	1.3077	1.52	1.52	5	5	2.31	196.91	51.21	110.5	550.0	215.4	12	125	OK	0.03	10.48	0.01	0.14	295.44	3600	0.15	1.12	OK
B-2	485.666	323.777	356.16	2.74	1.6553	1.68	1.68	5.5	5.5	2.82	258.11	95.88	143.8	550.0	405.2	12	125	OK	0.04	49.87	0.05	0.17	519.42	3600	0.26	1.12	OK
A-3	456.845	304.563	335.02	2.58	1.6062	1.68	1.68	5.5	5.5	2.82	242.8	90.19	139.5	550.0	380.9	12	125	OK	0.04	46.91	0.05	0.17	488.61	3600	0.25	1.12	OK
B-3	690.672	460.448	506.49	3.9	1.9748	1.98	1.98	6.5	6.5	3.92	264.26	173.8	178.3	550.0	738.8	12	125	OK	0.07	138.7	0.13	0.22	821.95	3600	0.42	1.12	OK
C-3	529.638	353.092	388.4	2.99	1.7292	1.83	1.83	6	6	3.35	237.23	118.9	153.4	550.0	503.1	12	125	OK	0.05	82.48	0.08	0.19	602.3	3600	0.3	1.12	OK
D-3	451.756	301.171	331.29	2.55	1.5969	1.68	1.68	5.5	5.5	2.82	240.09	89.19	138.7	550.0	376.6	12	125	OK	0.04	46.39	0.05	0.17	483.16	3600	0.24	1.12	OK
C-4	357.921	238.614	262.48	2.02	1.4213	1.52	1.52	5	5	2.31	232.38	60.44	120.0	550.0	254.6	12	125	OK	0.03	12.36	0.01	0.14	348.66	3600	0.18	1.12	OK
D-4	267.718	178.479	196.33	1.51	1.2288	1.52	1.52	5	5	2.31	173.81	45.21	103.8	550.0	190.0	12	125	OK	0.02	9.25	0.01	0.12	260.78	3600	0.13	1.12	OK
A-5	350.041	233.361	256.7	1.97	1.4036	1.52	1.52	5	5	2.31	227.26	59.11	118.7	550.0	248.9	12	125	OK	0.03	12.09	0.01	0.14	340.98	3600	0.17	1.12	OK
B-6	535.430	356.953	392.65	3.02	1.7378	1.83	1.83	6	6	3.35	239.82	120.2	154.2	550.0	508.7	12	125	OK	0.05	83.39	0.08	0.19	608.88	3600	0.31	1.12	OK
C-7	274.590	183.06	201.37	1.55	1.245	1.52	1.52	5	5	2.31	178.27	46.37	105.1	550.0	195.0	12	125	OK	0.02	9.48	0.01	0.12	267.48	3600	0.14	1.12	OK

DESIGN OF COMBINE FOOTING (Grid C-4/C-7)

Axial Factored loading by software

Column	C1(C4)	C2(C7)
Axial loading	357.921	274.59

Axial load on Column A=	238.61	kN
Axial load on Column B=	183.06	kN
Size of Column A=	355	mm
Size of Column B=	355	mm
Bearing Capacity of soil=	130	kN/sq.m

c/c dist.	1.4224	
left edge	0.1778	
edge to c/c dist	1.6002	Column A
Right edge	0.762	Column B

i.Length and Width of the footing

Centre of gravity of load from the property line

$$X = (PA \cdot x + PB \cdot y) / (PA + PB)$$

$$0.795307871 \text{ m}$$

Length of Footing=

$$2 \cdot X$$

$$1.590615742 \text{ m}$$

Assume

$$2.26 \text{ m}$$

Assume Weight of footing and earth above it as 5% of total weight

$$\text{Total load on earth} = (PA + PB) \cdot 1.05$$

$$442.7535 \text{ kN}$$

$$\text{Footing Width} = \text{Total Load on earth} / (\text{length of footing} \cdot \text{Bearing Capacity})$$

$$1.506989449 \text{ m}$$

Adopt

$$1.68 \text{ m width of footing}$$

ii.Longitudinal Bending moment and shear

$$\text{Factored load on Column A} = 357.915 \text{ kN}$$

$$\text{Factored load on Column B} = 274.59 \text{ kN}$$

$$\text{Net upward Soil Pressure} = (Pau + Pbu) / (\text{length of footing} \cdot \text{width of Footing})$$

$$166.5889697 \text{ kN/sq.m}$$

$$\text{Net upward soil pressure per unit length} = 279.87 \text{ kN/m}$$

Maximum shear force at centre line of column A

$$V1 = \text{Netupward soil pressure per unit length} \cdot \text{c/c distance}$$

$$\text{Negative shear force} = -49.76079159 \text{ kN}$$

$$V2 = V1 + Pau$$

$$\text{Positive shear force} = 308.1542084 \text{ kN}$$

Maximum Shear force at centre line of Column B

$$V1 = \text{Netupward soil pressure per unit length} \cdot \text{c/c distance}$$

$$\text{Negative shear force} = -61.3294646 \text{ kN}$$

$$V2 = V1 - Pbu$$

$$\text{Positive shear force} = 213.2605354 \text{ kN}$$

Point of Zero Shear force from centre of column A(from Calculation)

$$x = 1.186300175$$

Maximum bending moment computed from left side

$$BM = (\text{Net upward soil pressure (dist. From left edgt to A column} + X)^2) / 2 - 1.5 \cdot \text{Factored load} \cdot X$$

$$-164.2083709 \text{ KNm}$$

Maximum bending moment computed from Right side

$$BM = (\text{Net upward soil pressure (dist. From right edgt to B column} + X)^2) / 2 - 1.5 \cdot \text{Factored load B} \cdot X$$

$$205.4283955 \text{ KNm}$$

For computation thickness of footing

$$Bm = 0.138 \cdot fck \cdot b \cdot d^2$$

$$d^2 = 47218.87822$$

$$d = 217.2990525 \text{ mm}$$

$$400 \text{ mm}$$

Adopt effective depth as 400 mm & Overall depth as 450mm

iii) Main negative longitudinal reinforcement

$$BM = 0.87 \cdot f_y \cdot A_{st} (d - f_y \cdot a_{st} / (f_{ck} \cdot b))$$

Simplyfying this into Quadratic equation ($Ax^2 + Bx + c = 0$)

$$A_{st} = 1195.90 \text{ sqmm}$$

Ast	1195.90	mm ²
diameter of steel	12.00	mm
area of each steel	113.10	
spacing required	158.88	mm
<i>spacing provided</i>	150.00	mm

Development length for Φ mm bar=56 Φ

Sample Design of Slab (C/D/1/3)

Design Data

$$f_{ck} = 20 \text{ N/mm}^2$$

$$f_y = 500 \text{ N/mm}^2$$

Dimensions of the slab (c/c distance betⁿ supports),

$$\text{Length of short span, } l_x = 3.302 \text{ m}$$

$$\text{Length of long span, } l_y = 4.343 \text{ m}$$

$$\text{Width of the supporting beam,} = 230 \text{ mm}$$

$$\text{Clear cover to main reinforcement} = 15 \text{ mm}$$

$$\text{Assume dia. of reinforcement steel} = 10 \text{ mm}$$

Calculations

$$\text{Assume the thickness of slab as } 125 \text{ mm ;}$$

$$\text{Effective depth, } d_{\text{eff}} = 105 \text{ mm}$$

$$\text{Effective span, } l_x = 3.302 \text{ m (or) } 3.177 \text{ m whichever is less; } = 3.177 \text{ m}$$

$$l_y = 4.3434 \text{ m (or) } 4.218 \text{ m whichever is less; } = 4.218 \text{ m}$$

$$(l_y / l_x) = 1.33 < 2 ; \text{Here, } (l_y / l_x) \text{ is less than } 2$$

Hence design the slab as two way slab

Load Calculations

$$\text{Dead Load of slab} = 0.125 \times 25 = 3.13 \text{ KN/m}^2$$

$$\text{Finishing plus partition wall load on slab} = 1.00 \text{ KN/m}^2$$

$$\text{Total Dead load acting on the slab} = 4.13 \text{ KN/m}^2$$

$$\text{Live Load on slab} = 2.0 \text{ KN/m}^2$$

$$\text{Factored Design Load, } W = 9.20 \text{ KN/m}^2$$

Support Condition

(Type of panel according to support condition)

Two Adjacent Edges Discontinuous

For this support condition,

$$\text{Short span coefficient for } (l_y / l_x) = 1.33,$$

$$\text{Long span coefficient,}$$

$$\text{For negative moment, } a_x = 0.0668$$

$$\text{For negative moment, } a_y = 0.047$$

$$\text{For positive moment, } a_x = 0.0502$$

$$\text{For positive moment, } a_y = 0.035$$

Moment Calculation

$$\text{Max. BM per unit width, } M_x = a_x w l_x^2 \quad \& \quad M_y = a_y w l_x^2$$

	M_u KNm	M_u / bd^2 N/mm ²	P_t %	$A_{st, \text{req}}$ mm ²
For Short Span,				
At mid span,	4.66	0.42	0.10	104
At supports,	6.20	0.56	0.13	140
For Long span,				
At mid span,	3.25	0.36	0.08	89
At supports,	4.36	0.48	0.11	119

$$A_{st, \min} = (0.12/100) bD = 150 \text{ mm}^2$$

Reinforcement details

Provide 8 @ 150 mm c/c at midspan & supports for short span

$$(A_{st \text{ pro.}} = 335 \text{ mm}^2)$$

Provide 8 @ 150 mm c/c at midspan & supports for long span

$$(A_{st \text{ pro.}} = 335 \text{ mm}^2)$$

Check for Deflection

$$\text{Percentage of tension reinforcement} = 0.27 \%$$

$$f_s = 0.58 f_y (A_{st \text{ req}} / A_{st \text{ pro}}) = 90$$

Refer Fig. 4 of IS 456,

$$\text{Modification factor} = 2$$

$$(1/d)_{\max} = (1/d)_{\text{basic}} \times k_t \times k_c = 52$$

$$(1/d)_{\text{provided}} = 30$$

$$< (1/d)_{\max} \text{ (i.e. = 52)} \quad \text{OK}$$

Hence safe in deflection.

Check for shear

$$\text{Shear force per } m \text{ strip } (W \cdot l_x / 2), V = 14.6142 \text{ KN/m}$$

$$\text{Shear stress, } (V/bd), \tau_v = 0.120 \text{ N/mm}^2$$

$$P_t = 0.27\%$$

Design Shear Strength of Concrete

$$\tau_c = 0.442 \text{ N/mm}^2 \quad \text{Refer Table 19, IS 456}$$

$$> \tau_v \text{ (i.e. = 0.12)} \quad \text{OK}$$

Since $\tau_c > \tau_v$, so safe in shear.

Design of Staircase

1 Material Properties

<i>f_y</i>	<i>500.00</i>	N/mm ²
<i>f_{ck}</i>	<i>20.00</i>	N/mm ²
<i>cover</i>	<i>20.00</i>	mm
<i>diameter of main bars</i>	<i>12.00</i>	mm ²
<i>diameter of distribution bars</i>	<i>10.00</i>	mm ²

<i>Tread</i>	<i>0.25</i>	m
<i>Rise</i>	<i>0.18</i>	m
<i>Thickness of waist slab</i>	<i>0.152</i>	m
<i>width of slab</i>	<i>1.07</i>	m
<i>width of landing</i>	<i>1.07</i>	m
<i>thickness of landing slab</i>	<i>0.13</i>	m
<i>span of landing A</i>	<i>1.07</i>	m
<i>span of flight</i>	<i>1.78</i>	m
<i>span of landing B</i>	<i>1.07</i>	m
<i>Total Span</i>	<i>3.91</i>	m

2 LOAD CALCULATION

2.1 LOAD ON WAIST SLAB

area of slab	0.05	m ²
area of step	0.02	m ²
Total area	0.07	m ²
DLper m	1.75	KN/m
DL	6.87	KN/m ²
FF	1.30	KN/m ²
LL	3.00	KN/m ²
total load per m square	11.17	KN/m ²
Factored load	16.76	KN/m ²
load per meter	17.88	KN/m

2.2 LOAD ON LANDING

Self weight of slab	3.18	KN/m ²
FF	1.00	KN/m ²
LL	3.00	KN/m ²
load per meter square	7.18	KN/m ²
Factored load	10.76	KN/m ²
load per meter	11.48	KN/m

3 CALCULATION OF MOMENTS

reaction at B	28.14	m
Total load	56.29	m
reaction at A	28.14	m
		m
point of zero shear force		m
assuming it lies at the flight	1.96	m
assuming it lies at the landing	2.45	m
point of zero shear force	1.96	m
max moment at the span	30.56	KNm
design moment	30.56	KNm

4 DEPTH

effective depth from moment consideration	0.10	m
<i>overall depth provided</i>	<i>0.152</i>	m
effective depth	0.13	m

5 CALCULATION OF MAIN STEEL

A	10.19
B	-54810.00
C	30555018.90

Ast	631.69	mm ²
diameter of steel	12.00	mm
area of each steel	113.10	
spacing required	191.00	mm
<i>spacing provided</i>	<i>125.00</i>	mm
actual steel	965.22	mm ²

CALCULATION OF DISTRIBUTION STEEL

Ast	194.58	mm ²
diameter of steel	10.00	mm
area of each steel	78.54	
spacing required	430.59	mm
<i>spacing provided</i>	<i>175.00</i>	mm
actual steel	478.78	mm ²

CHECK FOR SHEAR

Vu	28.14
Tv	0.21
Pt	0.72
Tc	0.56
check	safe in shear

CHECK FOR DEVELOPMENT LENGTH

L_d in terms of ϕ	48.55	
M_1 (at A)	30.56	kNm
V (at A)	28.14	kN
L_o (anchorage required (A))	126.00	mm
L_d	1211.70	mm
$\phi <$	24.96	mm
Result	Safe	

7 DEFLECTION CHECK

Length (L) (m)	3.912
Breadth (b) (mm)	1066.8
Depth (D) (mm)	152.4
d (mm)	126.00
f_{ck} (N/mm ²)	20
f_y (N/mm ²)	500
Moment (M) KNm	30.555
M_u/bd^2	1.804
$A_{st}(reqd)$ mm ²	631.686
$A_{st}(prov)$ mm ²	965.218
α	23
β	1
γ	1.460
δ	1
λ	1
$\alpha\beta\gamma\delta\lambda$	33.574
Reqd (d)	116.505
L/d	31.044
Check	ok

Design of Strap Beam

Grade of steel, f_y :	500	N/mm ²			
Concrete grade, f_{ck} :	20	N/mm ²			
Soil bearing capacity, q_o :	130	KN/m ²			
Column C1			Column C2		
Width:	355	mm	Width:	355	mm
Depth:	355	mm	Depth:	355	mm
Factored load, P_{u1} :	529.638	KN	Factored load, P_{u1} :	451.756	KN

Size of footing

C/C spacing between column, L:	3.302	m	
C.G. of loading, x:	1.520	m	[From C2]
Area of footing required:	5.536	m ²	[Assuming 10% self weight of the footing]
Width:	1.830	m	[Assumed]
L1:	2.329	m	
L2:	0.696	m	

Design of slab footing under columns

Soil pressure:	177.27	KN/m ²	
Width of strap beam:	0.35	m	[Assumed]
Max moment at the face of strap beam:	48.54	KN-m per m run	
Depth required from moment criteria:	133	mm	
Effective depth adopted:	500	mm	
Overall depth adopted:	550	mm	
Reinforcement in footing slab			
A_{st} , required	754	mm ²	
Provide 12 mm ϕ bars @ 150 mm c/c			
A_{st} , provided	754	mm ²	
P_t	0.151	%	

Check for one way shear criteria

Factored shear force, V_{u1} :	42.5	KN	
Nominal shear stress, t_v :	0.085	N/mm ²	
Design shear strength of concrete, t_c :	0.29	N/mm ²	
t_v	>	t_c	OK

Design of Strap beam

Width of strap beam:	350	mm	
Maximum moment:	338.34	KN-m	
Effective depth required for moment criteria:	592	mm	
Effective depth adopted:	550	mm	
Overall depth adopted:	600	mm	
Negative reinforcement			
Maximum negative moment	338.34	KN-m	
Area of steel required:	1722	mm ²	
Provide 6 no.s of 20 mm ϕ bars			
Area of steel provided:	1885	mm ²	
p_t :	0.9792	%	

Positive reinforcement

Maximum positive moment: 19.65 KN-m

Area of steel required: 83 mm²

Provide 5 no.s of 16 mm ϕ bars

Area of steel provided: 1005 mm²

pt: 0.522 %

Design for shear

Factored shear force, V_u = 281 KN

Nominal shear stress, t_v = 1.461 N/mm²

Design shear strength of concrete, t_c = 0.497 N/mm²

Factored shear force, V_{us} = 185.57 KN

Provide 2 legged 10 mm ϕ bars @ 200 mm c/c at support

12. Detailing of the Structural Elements:

The Reinforcement detailing of most of the important structural components have been shown in drawing. They confirm with the relevant sections of the IS Codes IS 456-2000, NBC 105: 2077, SP-16, and NBC 205-2013.

13. ANNEX

ANNEX

Story Response - Auto Lateral Load to Stories

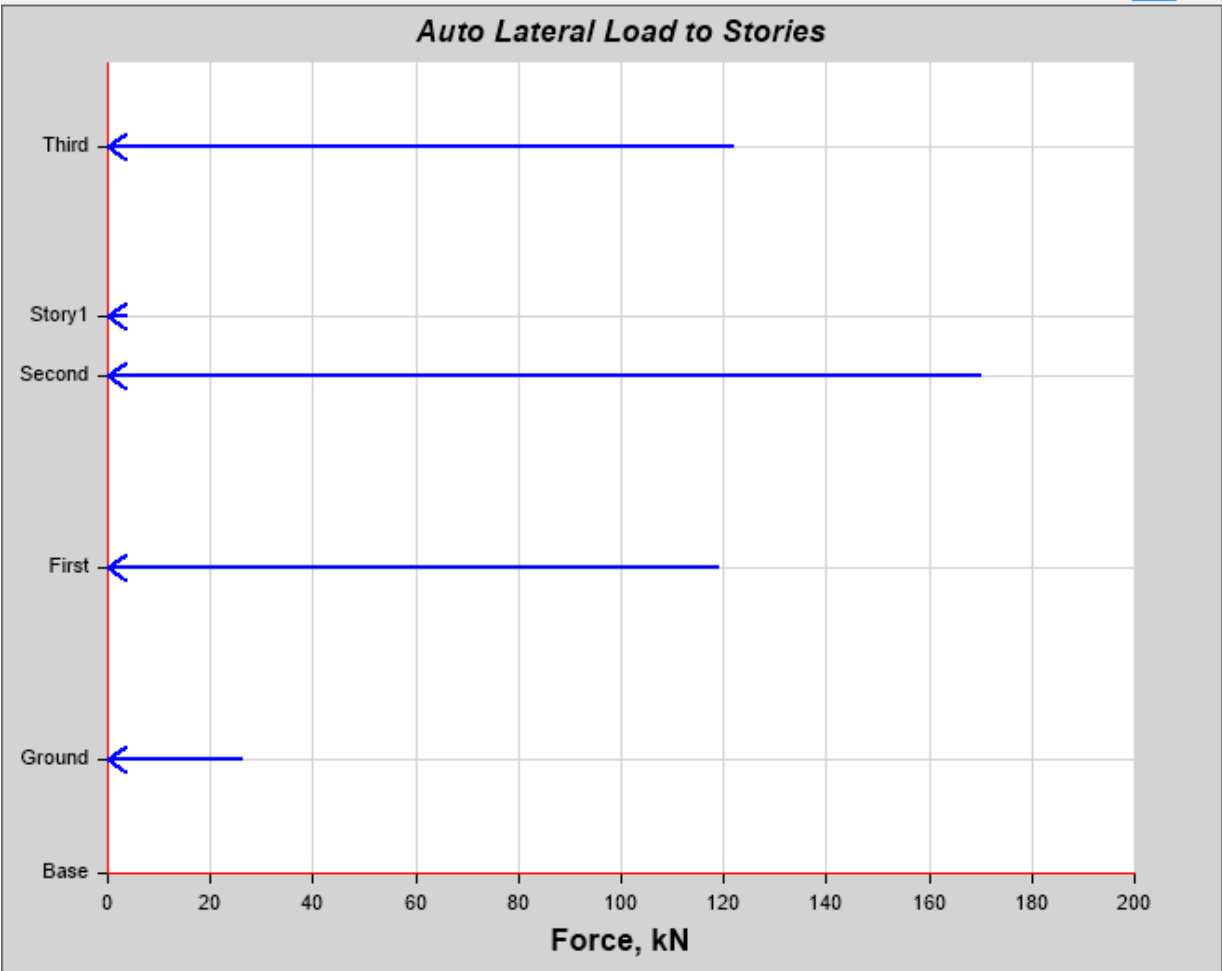
14. Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

15. Input Data

Name	StoryResp1		
Display Type	Auto lateral loads to stories	Story Range	All Stories
Load Pattern	EQxULTIMATE	Top Story	Third
Load Set	1	Bottom Story	Base

16. Plot



Story Response - Auto Lateral Load to Stories

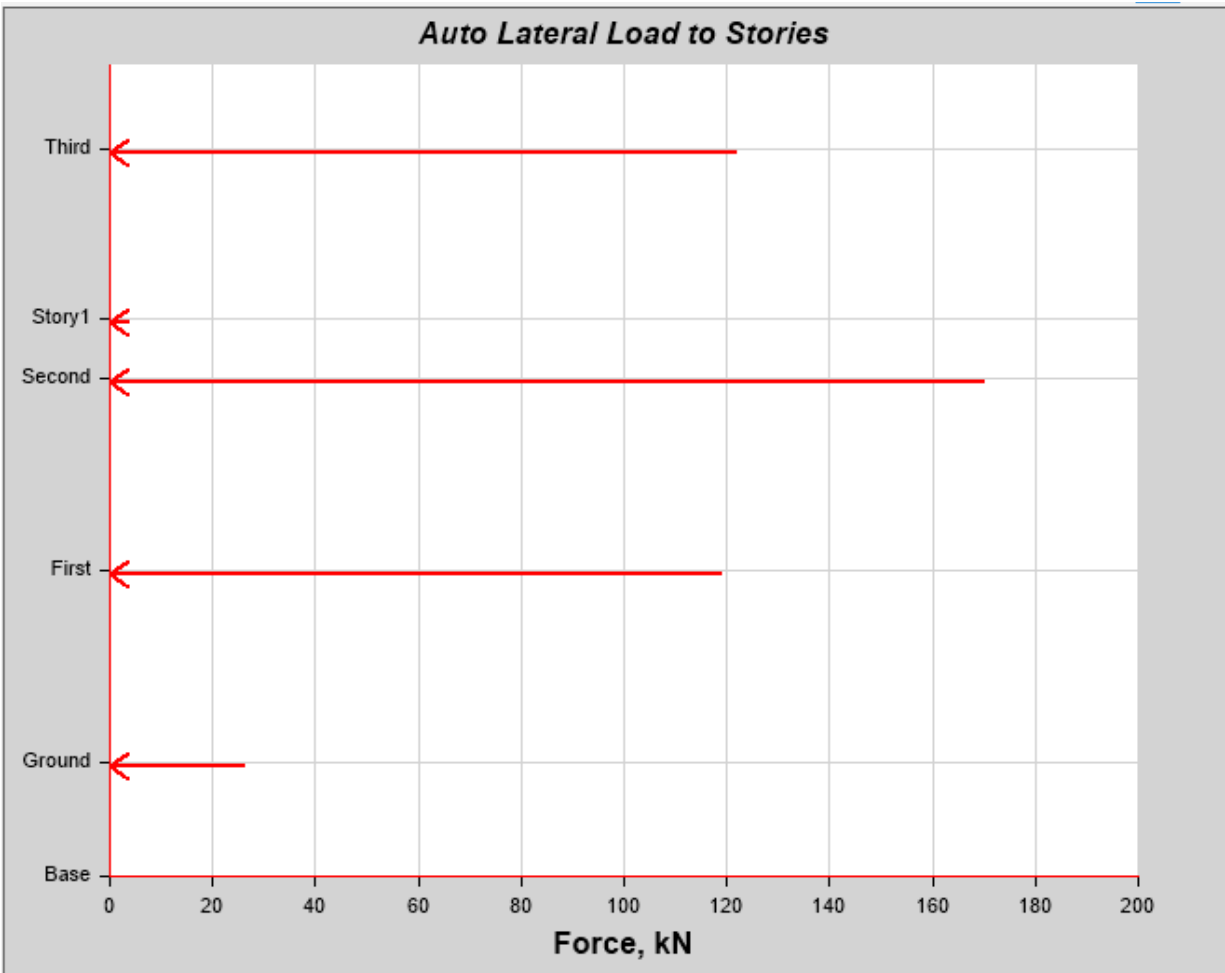
17. Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

18. Input Data

Name	StoryResp1		
Display Type	Auto lateral loads to stories	Story Range	All Stories
Load Pattern	EQyULTIMATE	Top Story	Third
Load Set	1	Bottom Story	Base

19. Plot



Story Response - Auto Lateral Load to Stories

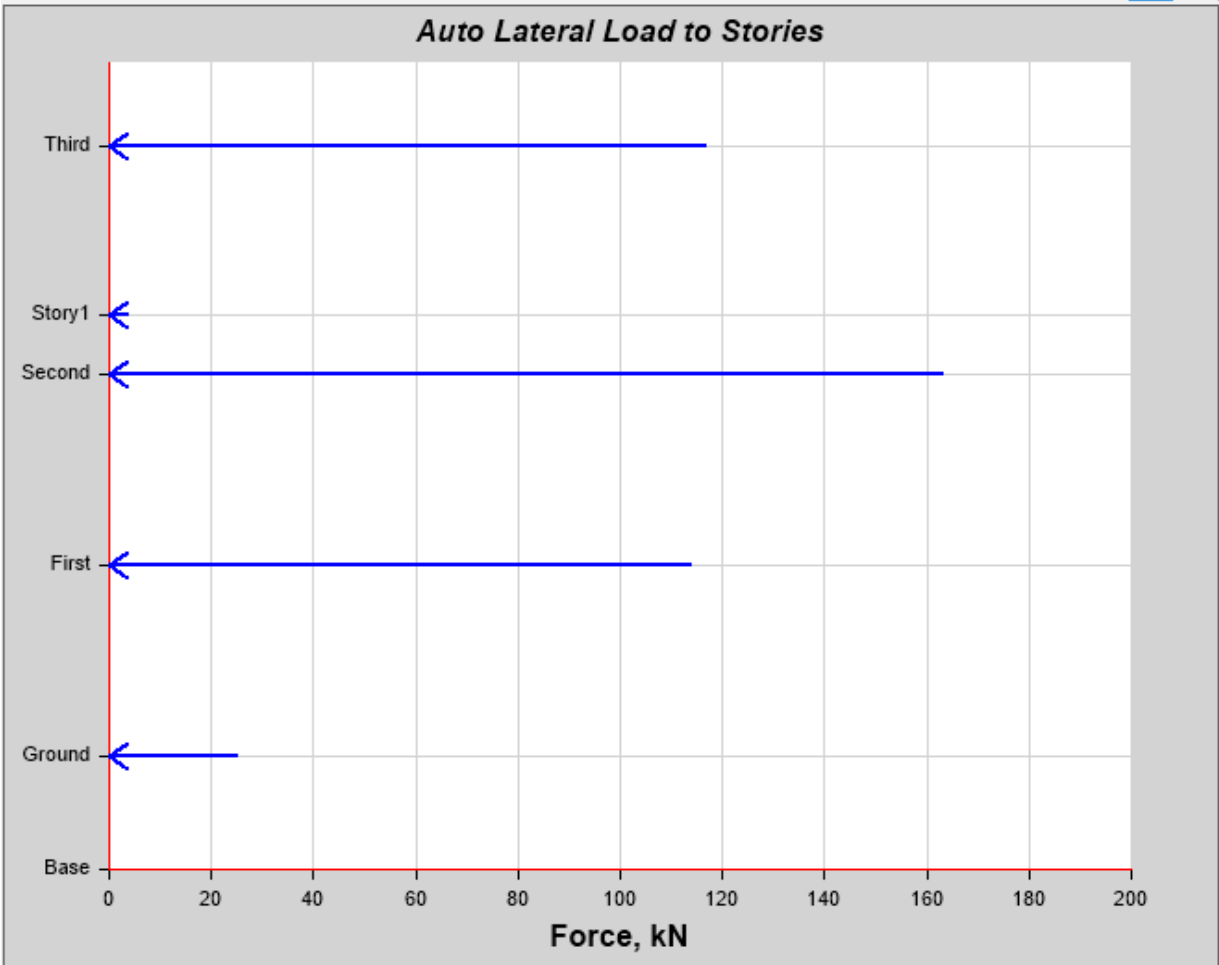
20. Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

21. Input Data

Name	StoryResp1		
Display Type	Auto lateral loads to stories	Story Range	All Stories
Load Pattern	EQxSERVICE	Top Story	Third
Load Set	1	Bottom Story	Base

22. Plot



Story Response - Auto Lateral Load to Stories

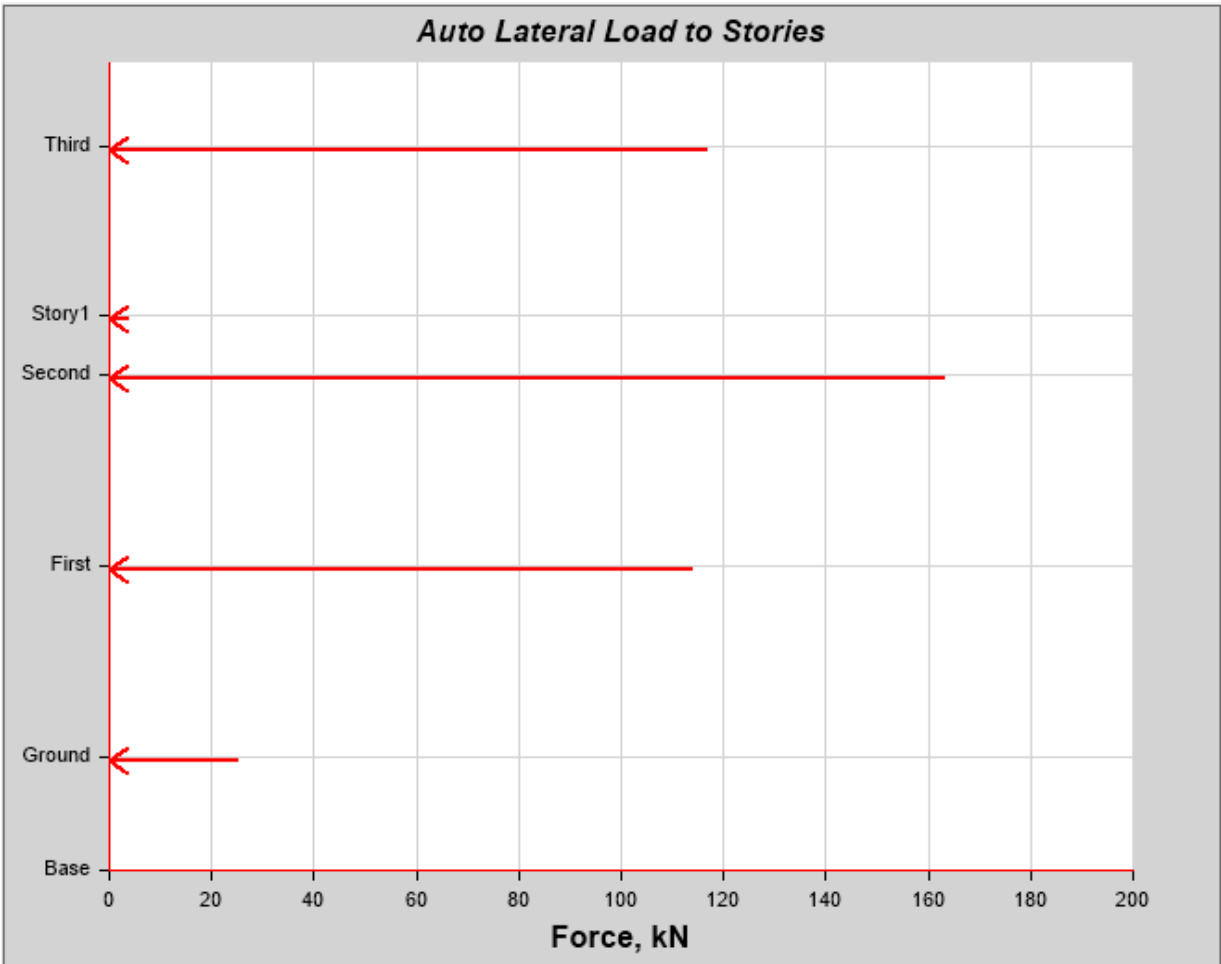
23. Summary Description

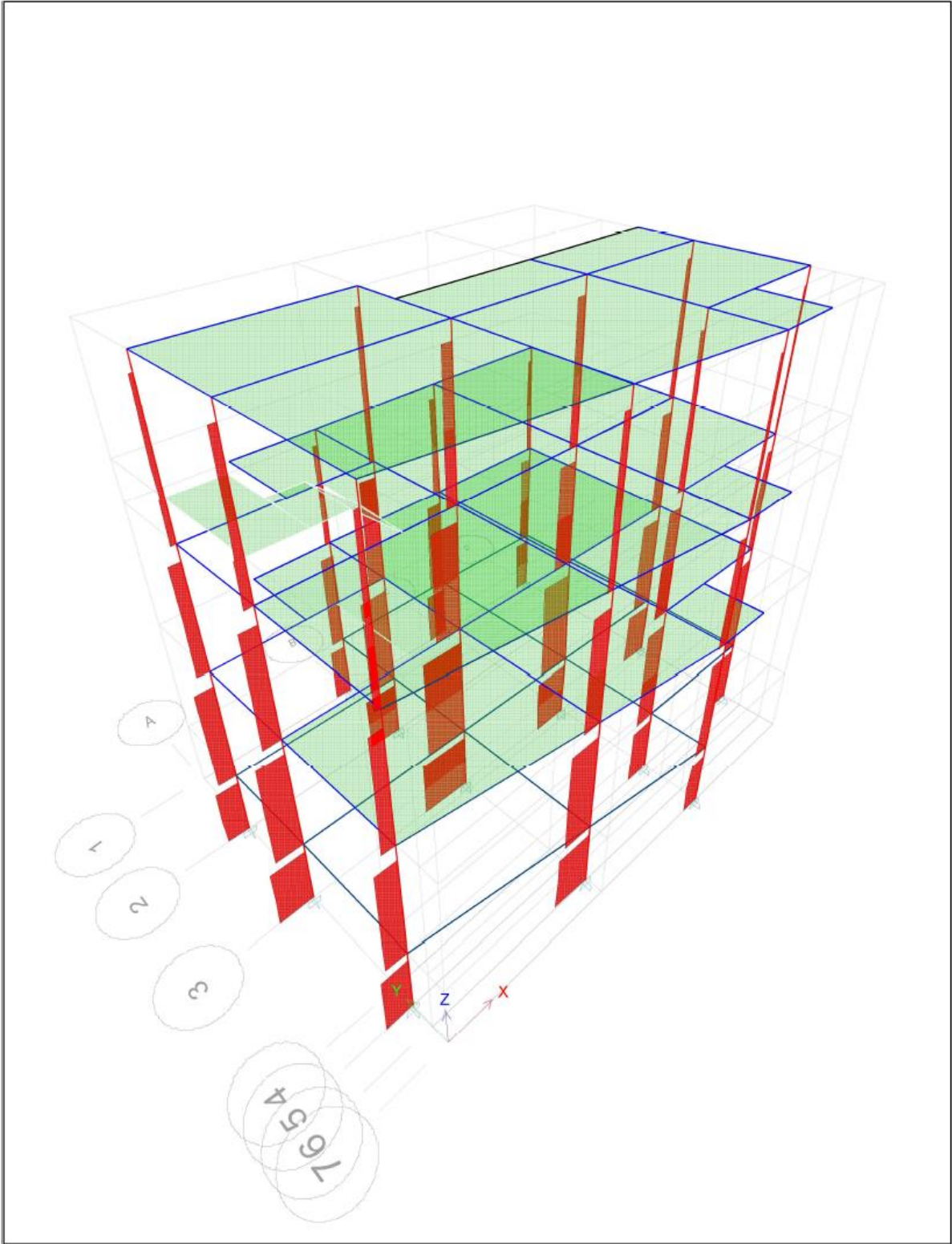
This is story response output for a specified range of stories and a selected load case or load combination.

24. Input Data

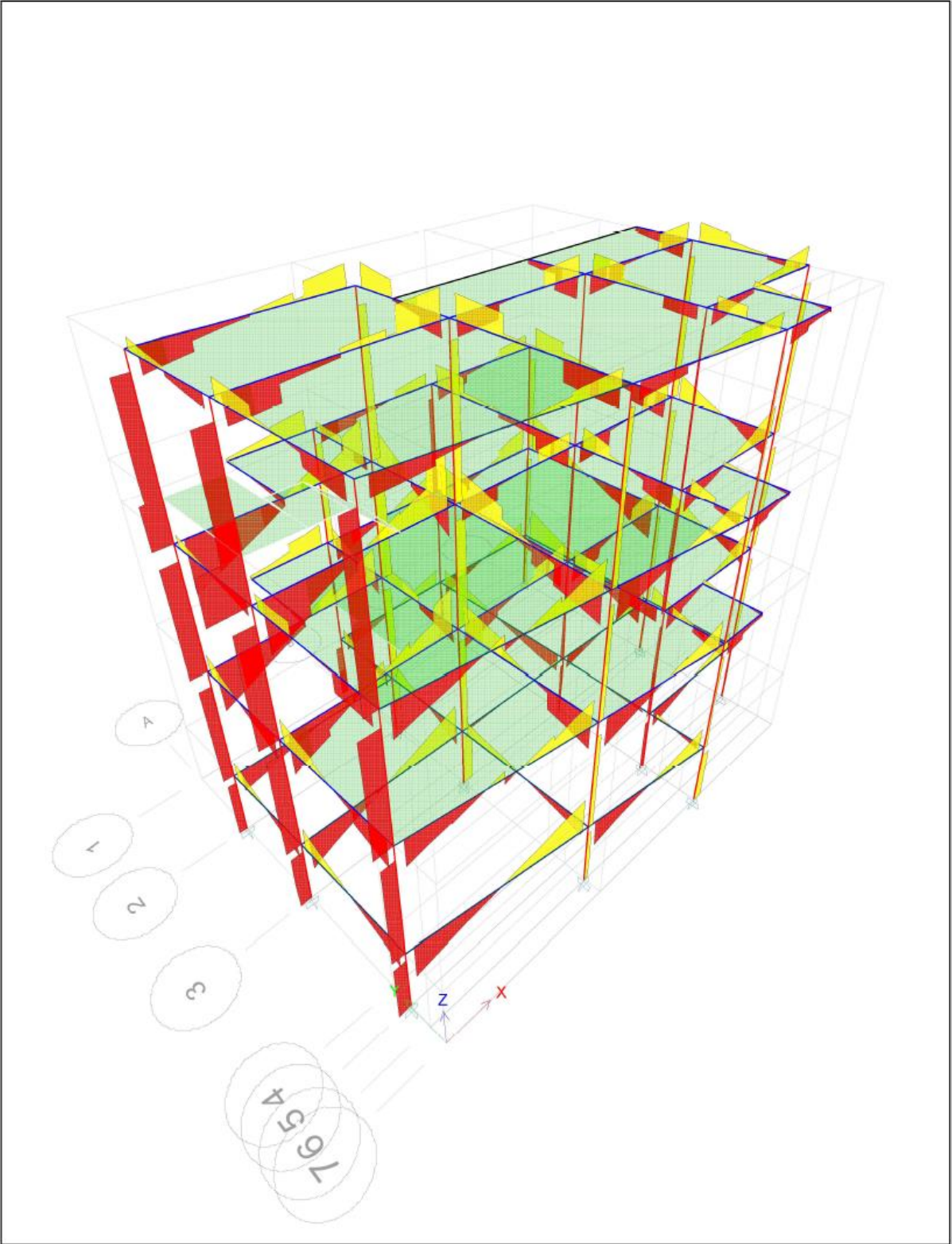
Name	StoryResp1		
Display Type	Auto lateral loads to stories	Story Range	All Stories
Load Pattern	EQySERVICE	Top Story	Third
Load Set	1	Bottom Story	Base

25. Plot

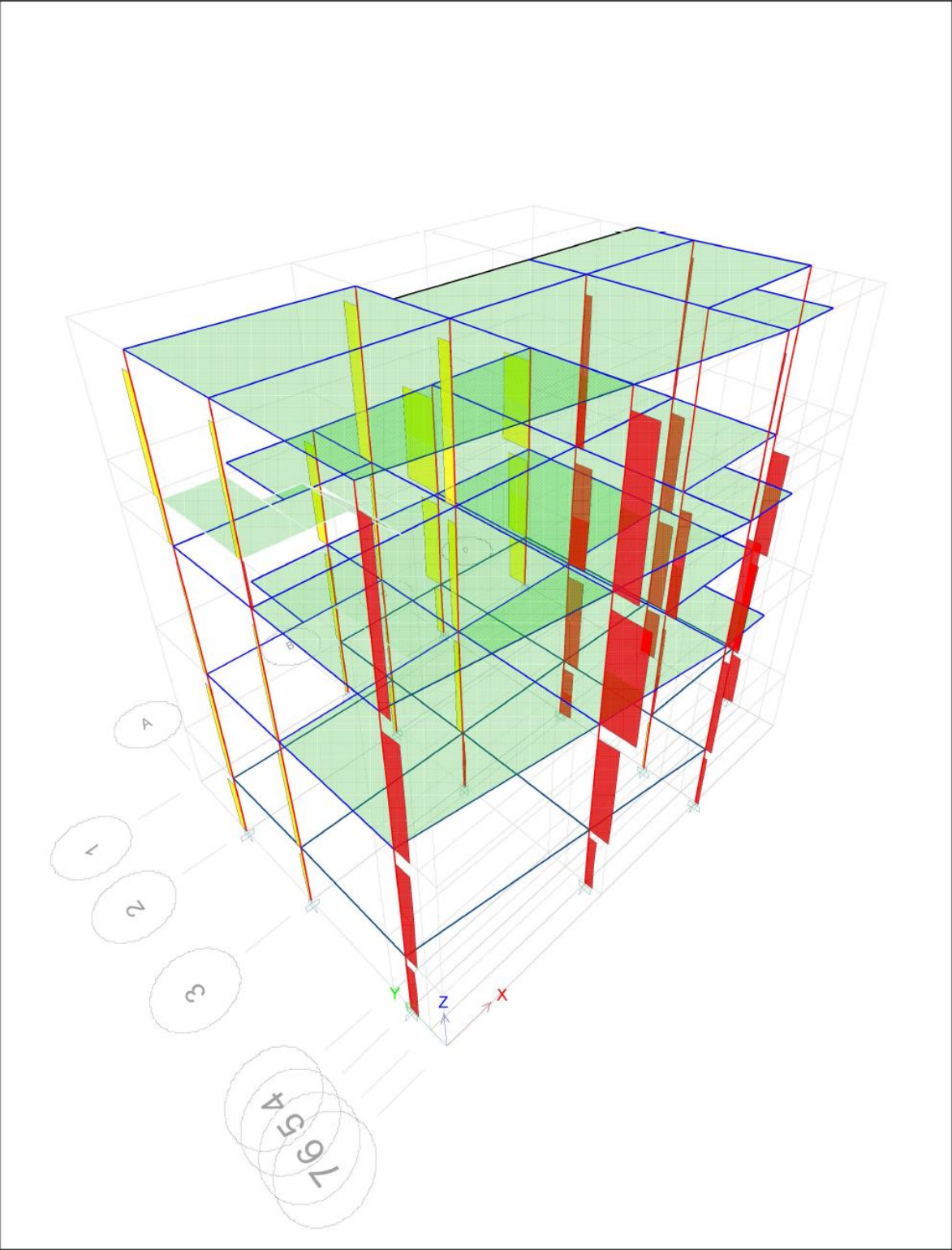




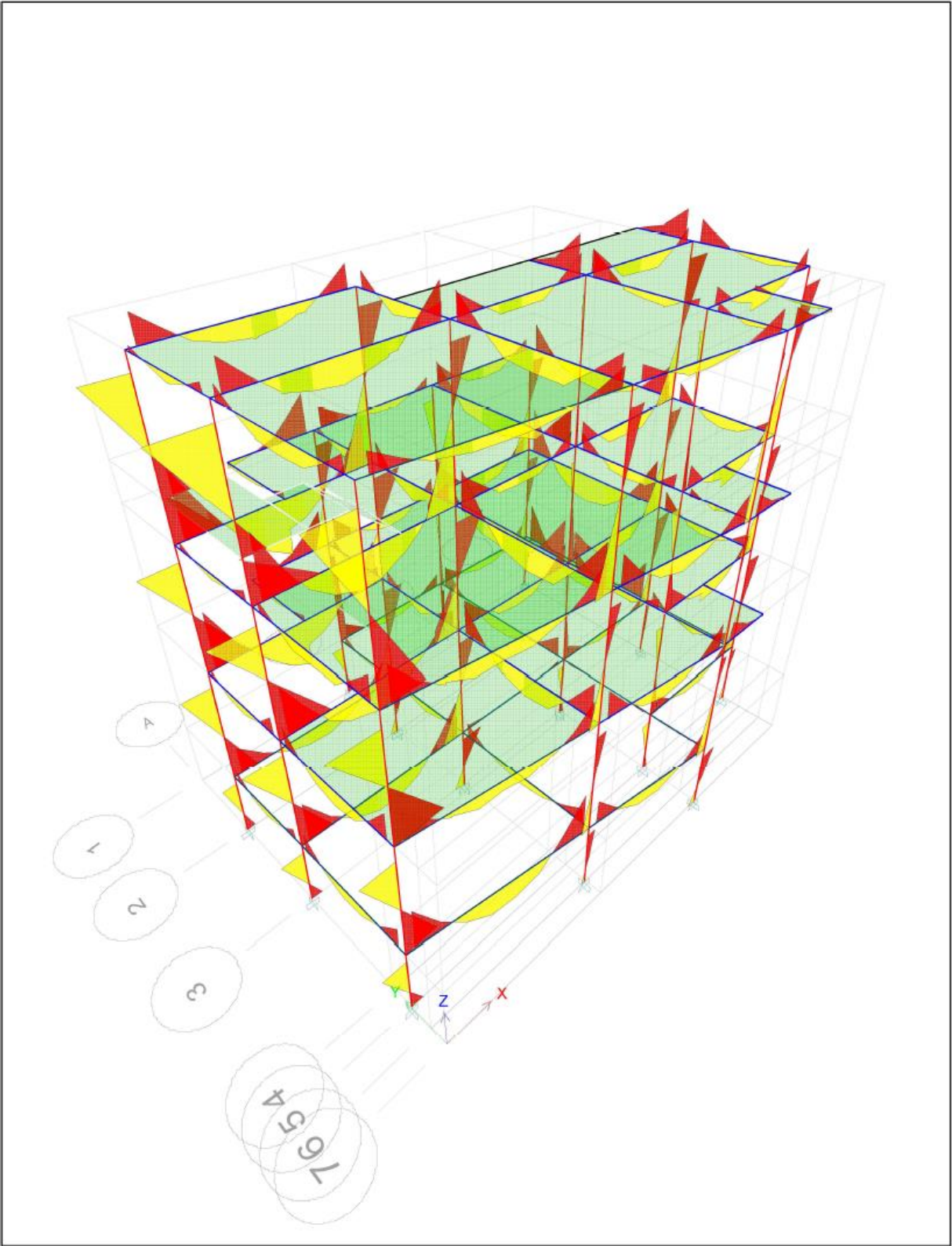
RunModel.EDB 3-D View Axial Force Diagram (1.5DL+1.5LL) [kN]



RunModel.EDB 3-D View Shear Force 2-2 Diagram (1.5DL+1.5LL) [kN]



RunModel.EDB 3-D View Shear Force 3-3 Diagram (1.5DL+1.5LL) [kN]



RunModel.EDB 3-D View Moment 3-3 Diagram (1.5DL+1.5LL) [kN-m]

TABLE: Load Pattern Definitions - Auto Seismic - IS 1893 2002								
Name	Ecc Ratio	Top Story	Bottom Story	Period Type	C	K	Weight Used	Base Shear
							KN	KN
EQxSERVICE	0.1	Story4	Base	User Defined	0.126	1.028671		
EQxSERVICE(1/3)	0.1	Story4	Base	User Defined	0.126	1.028671	3365.2161	424.0172
EQxSERVICE(2/3)	0.1	Story4	Base	User Defined	0.126	1.028671	3365.2161	424.0172
EQxSERVICE(3/3)	0.1	Story4	Base	User Defined	0.126	1.028671	3365.2161	424.0172
EQxULTIMATE	0.1	Story4	Base	User Defined	0.13125	1.028671		
EQxULTIMATE(1/3)	0.1	Story4	Base	User Defined	0.13125	1.028671	3365.2161	441.6846
EQxULTIMATE(2/3)	0.1	Story4	Base	User Defined	0.13125	1.028671	3365.2161	441.6846
EQxULTIMATE(3/3)	0.1	Story4	Base	User Defined	0.13125	1.028671	3365.2161	441.6846
EQySERVICE	0.1	Story4	Base	User Defined	0.126	1.028671		
EQySERVICE(1/3)	0.1	Story4	Base	User Defined	0.126	1.028671	3365.2161	424.0172
EQySERVICE(2/3)	0.1	Story4	Base	User Defined	0.126	1.028671	3365.2161	424.0172
EQySERVICE(3/3)	0.1	Story4	Base	User Defined	0.126	1.028671	3365.2161	424.0172
EQyULTIMATE	0.1	Story4	Base	User Defined	0.13125	1.028671		
EQyULTIMATE(1/3)	0.1	Story4	Base	User Defined	0.13125	1.028671	3365.2161	441.6846
EQyULTIMATE(2/3)	0.1	Story4	Base	User Defined	0.13125	1.028671	3365.2161	441.6846

EARTHQUAKE SERVICE (EQ SERVICE)

TABLE: Story Displacement				
Story	Elevation	Location	X-Dir	Y-Dir
	m		mm	mm
Third	10.7696	Top	29.885	3.01
Story1	8.255	Top	23.388	2.68
Second	7.366	Top	23.965	2.537
First	4.5212	Top	15.548	1.434
Ground	1.6764	Top	3.836	0.313
Base	0	Top	0	0
Story	Elevation	Location	X-Dir	Y-Dir
	m		mm	mm
Third	10.7696	Top	1.301	24.979
Story1	8.255	Top	0.74	21.432
Second	7.366	Top	1.213	20.251
First	4.5212	Top	0.741	13.443
Ground	1.6764	Top	0.169	3.395
Base	0	Top	0	0

TABLE: Story Drift				
Story	Elevation	Location	X-Dir	Y-Dir
	m			
Third	10.7696	Top	0.00177	0.000139
Story1	8.255	Top	0.001412	0.00016
Second	7.366	Top	0.002959	0.000388
First	4.5212	Top	0.004117	0.000394
Ground	1.6764	Top	0.002288	0.000187
Base	0	Top	0	0
Story	Elevation	Location	X-Dir	Y-Dir
	m			
Third	10.7696	Top	0.000152	0.00141
Story1	8.255	Top	0.000312	0.001329
Second	7.366	Top	0.000166	0.002393
First	4.5212	Top	0.000203	0.003532
Ground	1.6764	Top	0.000101	0.002025
Base	0	Top	0	0

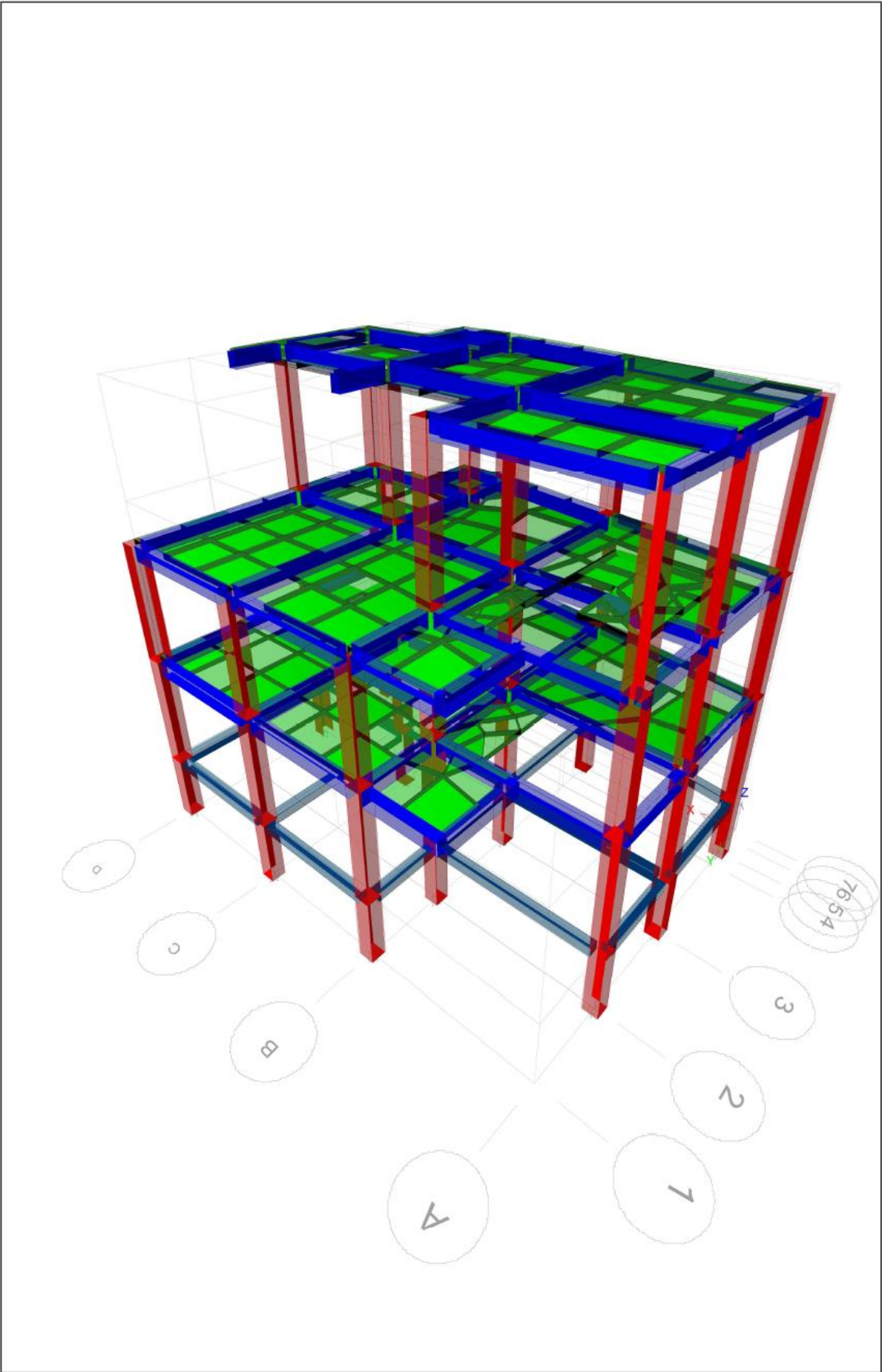
EARTHQUAKE ULTIMATE (EQ ULTIMATE)

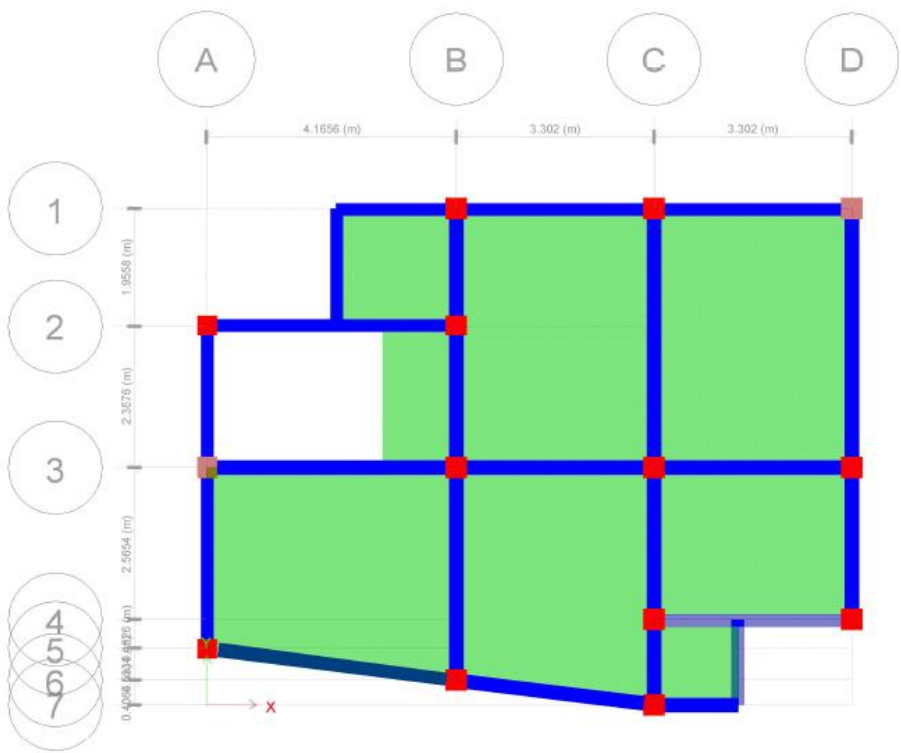
TABLE: Story Displacement				
Story	Elevation	Location	X-Dir	Y-Dir
	m		mm	mm
Third	10.7696	Top	31.13	3.135
Story1	8.255	Top	24.363	2.791
Second	7.366	Top	24.963	2.643
First	4.5212	Top	16.196	1.494
Ground	1.6764	Top	3.996	0.326
Base	0	Top	0	0
Story	Elevation	Location	X-Dir	Y-Dir
	m		mm	mm
Third	10.7696	Top	1.355	26.02
Story1	8.255	Top	0.771	22.325
Second	7.366	Top	1.263	21.095
First	4.5212	Top	0.772	14.004
Ground	1.6764	Top	0.177	3.536
Base	0	Top	0	0

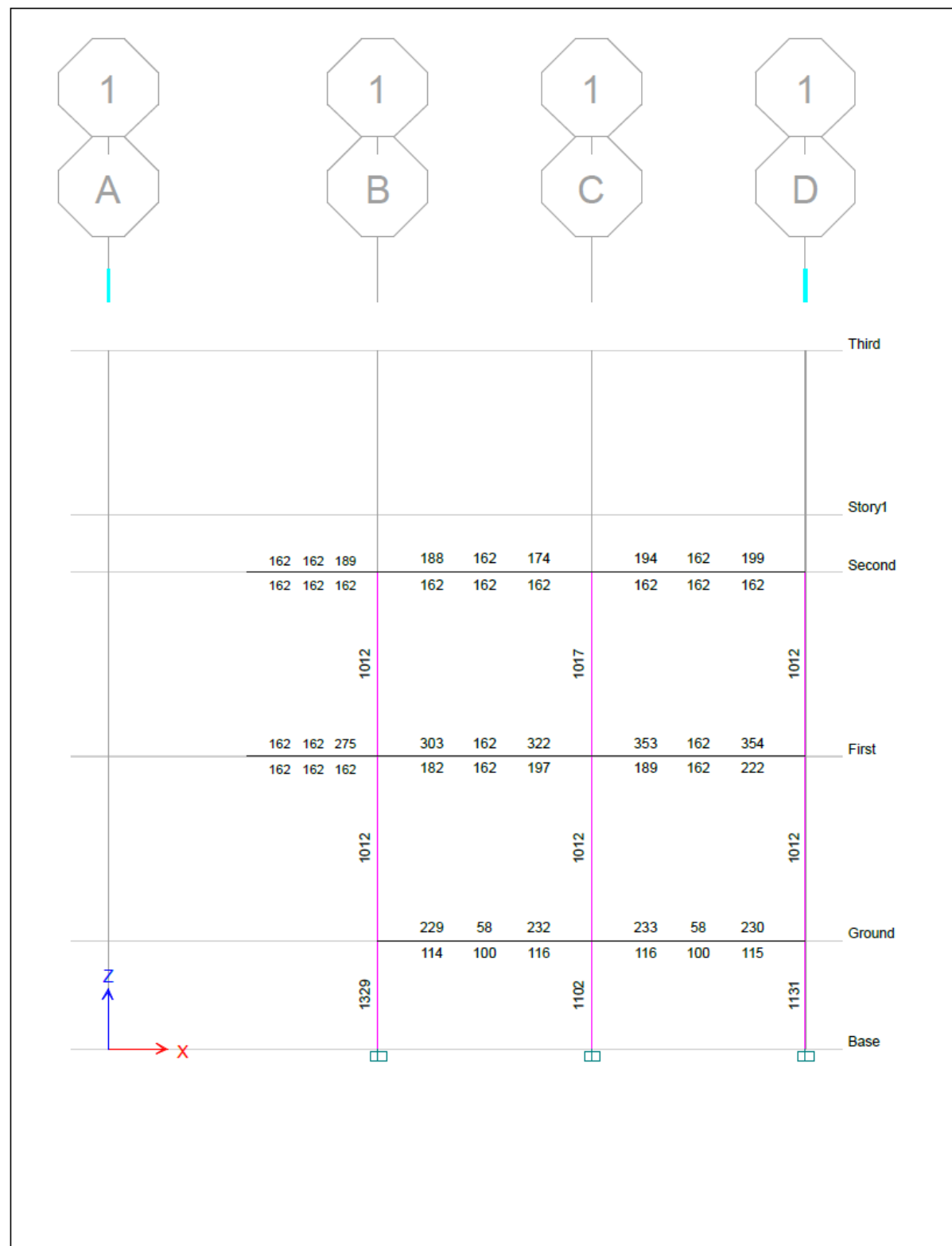
TABLE: Story Drift				
Story	Elevation	Location	X-Dir	Y-Dir
	m			
Third	10.7696	Top	0.001844	0.000144
Story1	8.255	Top	0.001471	0.000166
Second	7.366	Top	0.003082	0.000404
First	4.5212	Top	0.004289	0.00041
Ground	1.6764	Top	0.002383	0.000195
Base	0	Top	0	0
Story	Elevation	Location	X-Dir	Y-Dir
	m			
Third	10.7696	Top	0.000158	0.001469
Story1	8.255	Top	0.000325	0.001384
Second	7.366	Top	0.000173	0.002493
First	4.5212	Top	0.000212	0.00368
Ground	1.6764	Top	0.000105	0.002109
Base	0	Top	0	0

Soft Storey / Stiffness Irregularity check										
Story	Soft Storey Check						Extreme Soft Storey Check			
	X-direction			y-direction			X-direction		y-direction	
	Lateral Stiffness	Lateral Stiffness isn't less than 70% of that in the above storey	Lateral Stiffness isn't less than 80% of the average stiffness of 3 storeys above	Lateral Stiffness	Lateral Stiffness isn't less than 70% of that in the above storey	Lateral Stiffness isn't less than 80% of the average stiffness of 3 storeys above	Lateral Stiffness isn't less than 60% of that in the storey above	Lateral Stiffness isn't less than 70% of the average stiffness of 3 storey above	Lateral Stiffness isn't less than 60% of that in the storey above	Lateral Stiffness isn't less than 70% of the average stiffness of 3 storey above
Third	0	ok	ok	0	ok	ok	ok	ok	ok	ok
Story1	57664.746	ok	ok	57664.746	ok	ok	ok	ok	ok	ok
Second	37164.868	ok	ok	37164.868	ok	ok	ok	ok	ok	ok
First	36543.663	ok	ok	36543.663	ok	ok	ok	ok	ok	ok
Ground	117778.637	ok	ok	117778.637	ok	ok	ok	ok	ok	ok

TABLE: Modal Periods And Frequencies					
Case	Mode	Period	Frequency	CircFreq	Eigenvalue
		sec	cyc/sec	rad/sec	rad ² /sec ²
Modal	1	0.731	1.368	8.5974	73.9149
Modal	2	0.67	1.493	9.3823	88.0267
Modal	3	0.583	1.715	10.778	116.1657
Modal	4	0.233	4.289	26.9514	726.3788
Modal	5	0.221	4.525	28.4304	808.2865
Modal	6	0.201	4.98	31.2873	978.8974
Modal	7	0.127	7.87	49.4468	2444.9824
Modal	8	0.123	8.143	51.1648	2617.8321
Modal	9	0.112	8.89	55.8548	3119.7562
Modal	10	0.065	15.499	97.381	9483.0582
Modal	11	0.063	15.757	99.0035	9801.6873
Modal	12	0.061	16.472	103.4995	10712.1456

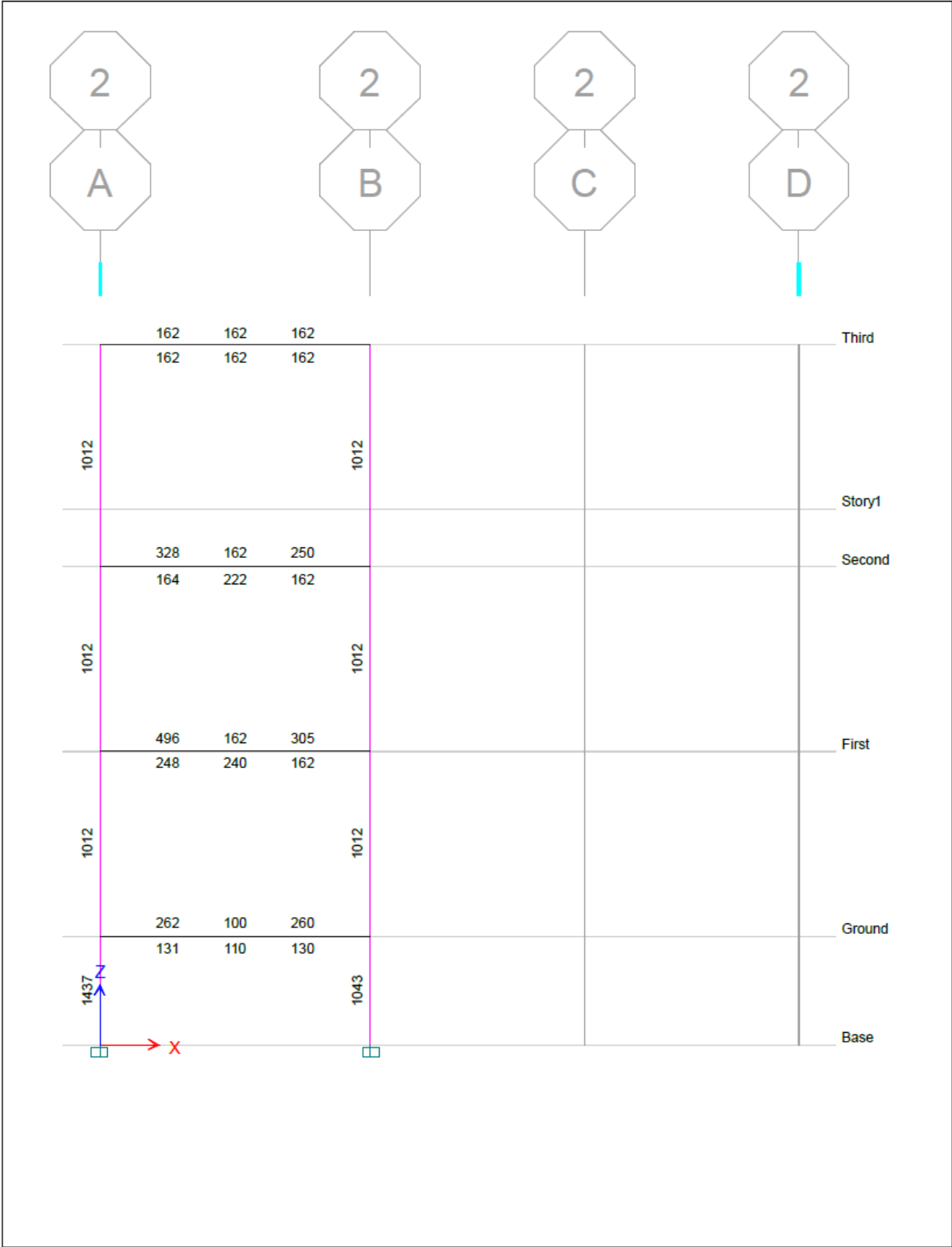






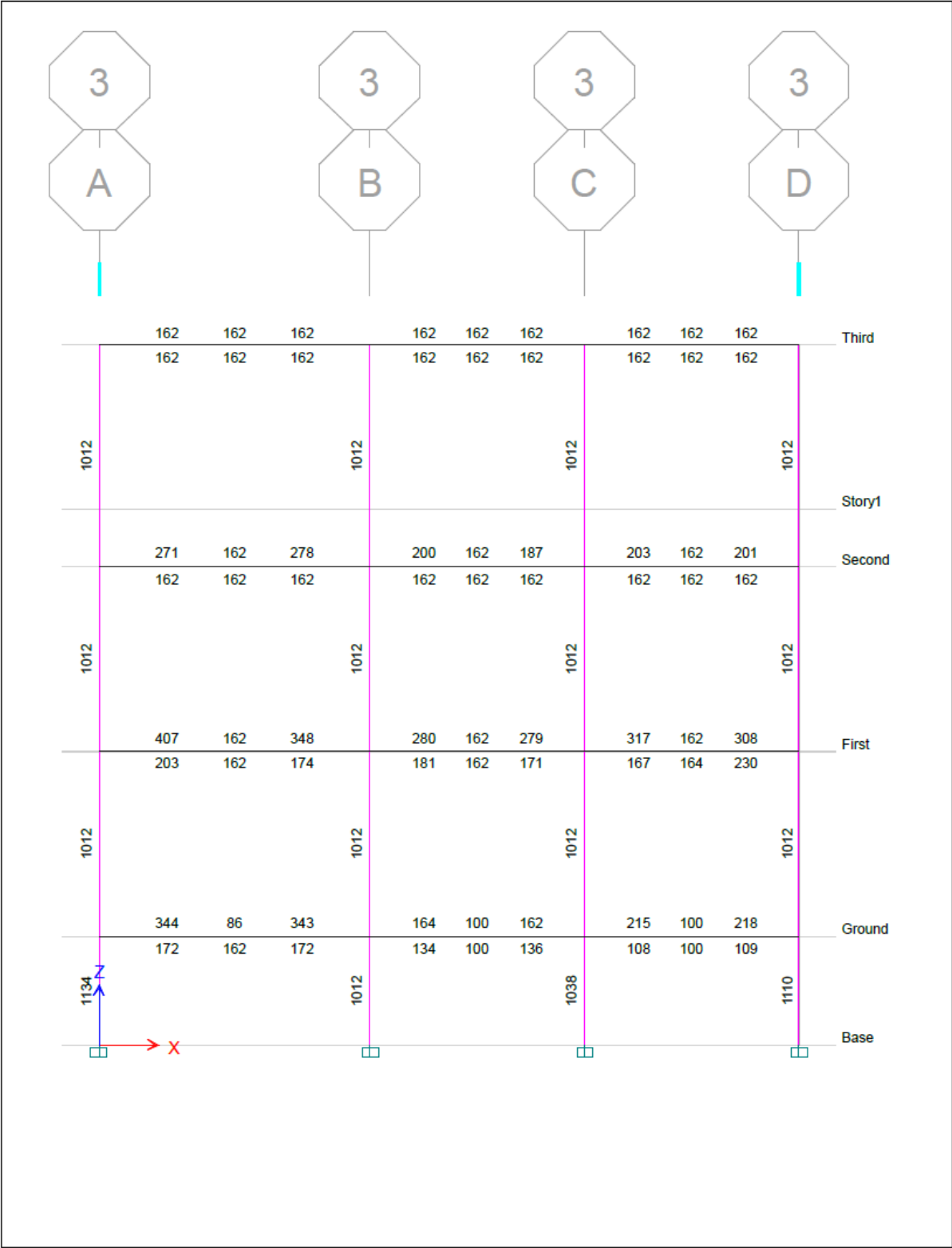
RunModel.EDB

Elevation View - 1 Longitudinal Reinforcing (IS 456:2000)



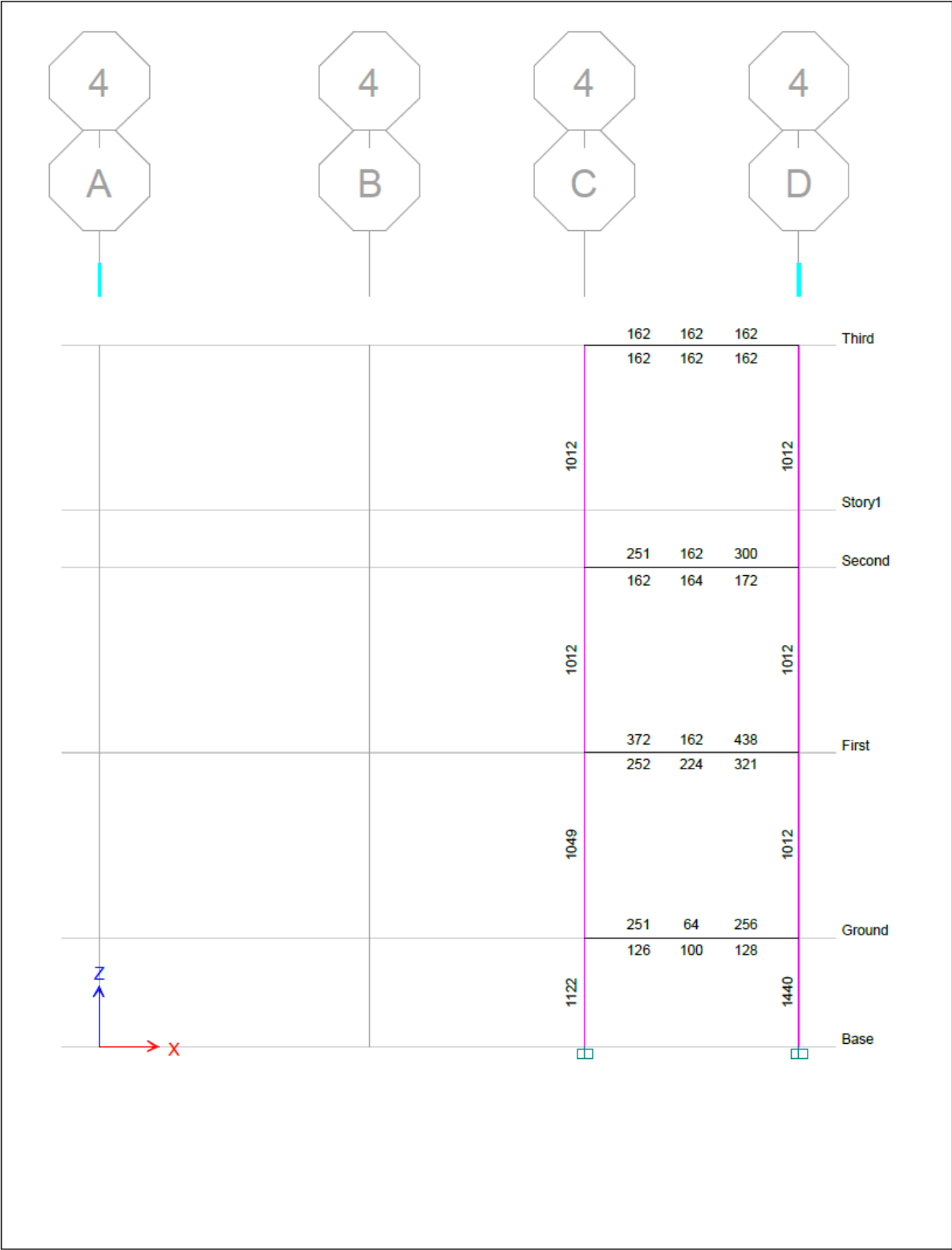
RunModel.EDB

Elevation View - 2 Longitudinal Reinforcing (IS 456:2000)



RunModel.EDB

Elevation View - 3 Longitudinal Reinforcing (IS 456:2000)

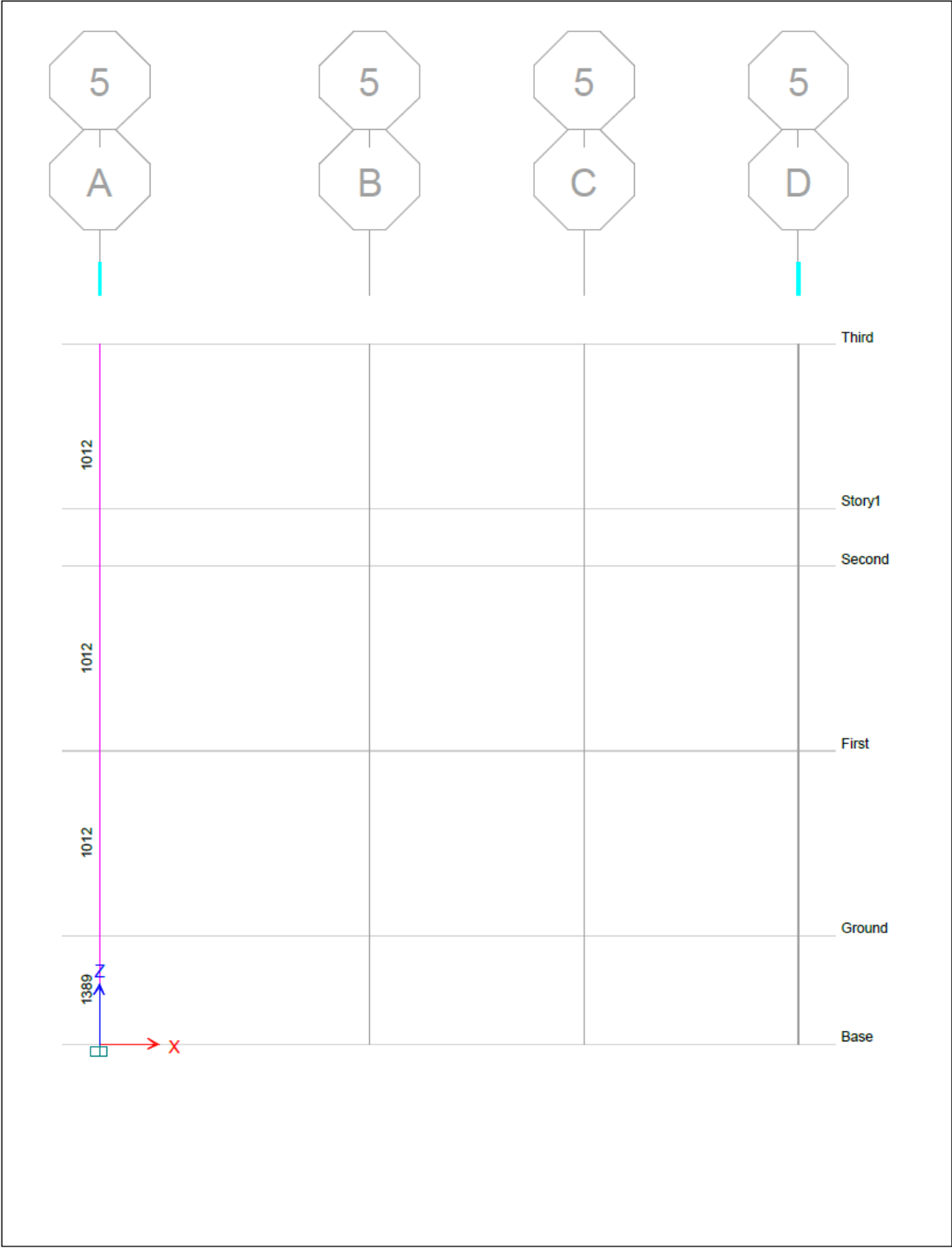


RunModel.EDB

Elevation View - 4 Longitudinal Reinforcing (IS 456:2000)

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RunModel.EDB

Elevation View - 5 Longitudinal Reinforcing (IS 456:2000)



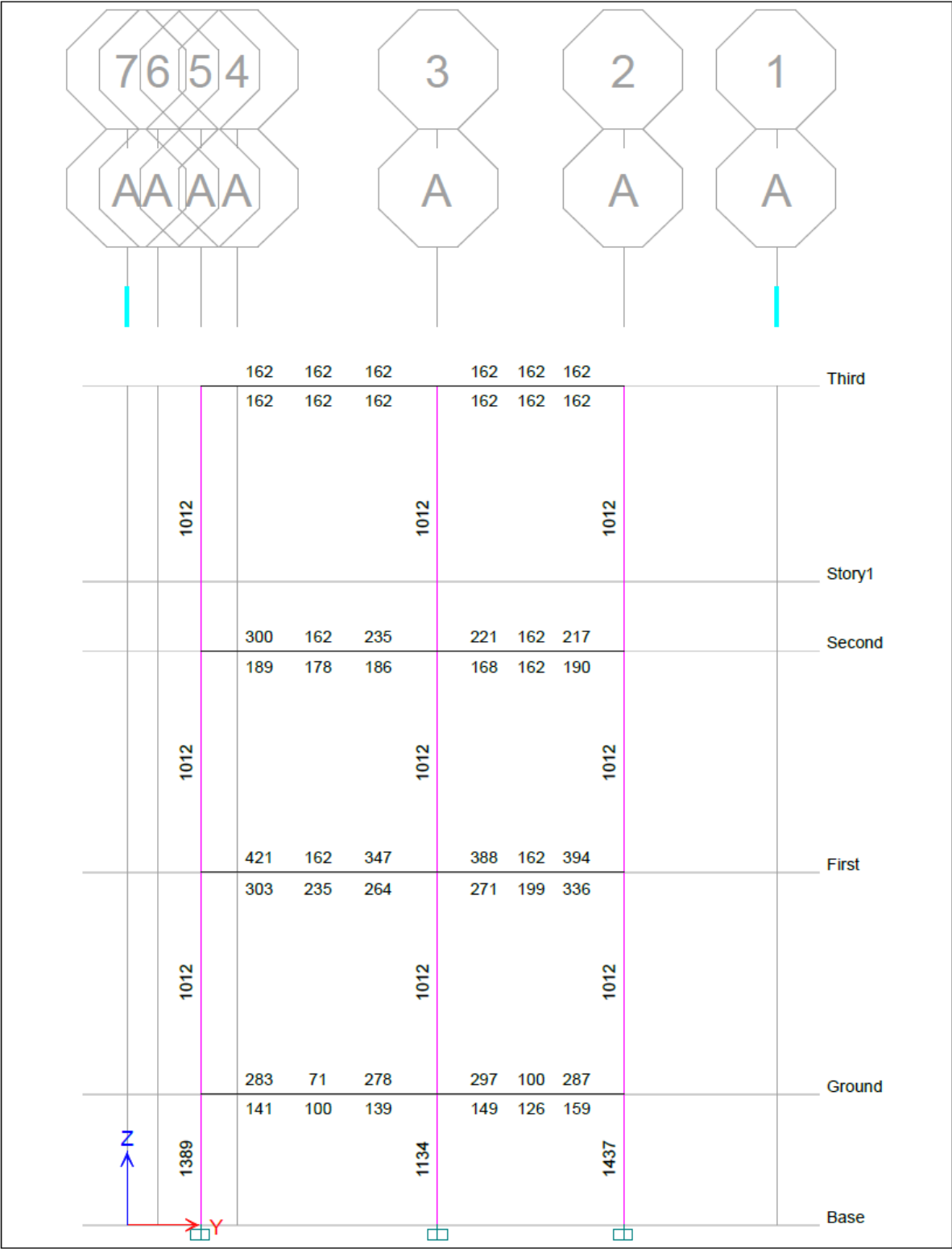
RunModel.EDB

Elevation View - 6 Longitudinal Reinforcing (IS 456:2000)



RunModel.EDB

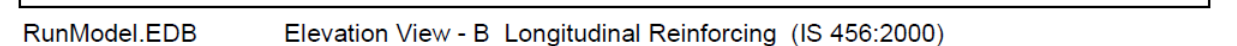
Elevation View - 7 Longitudinal Reinforcing (IS 456:2000)

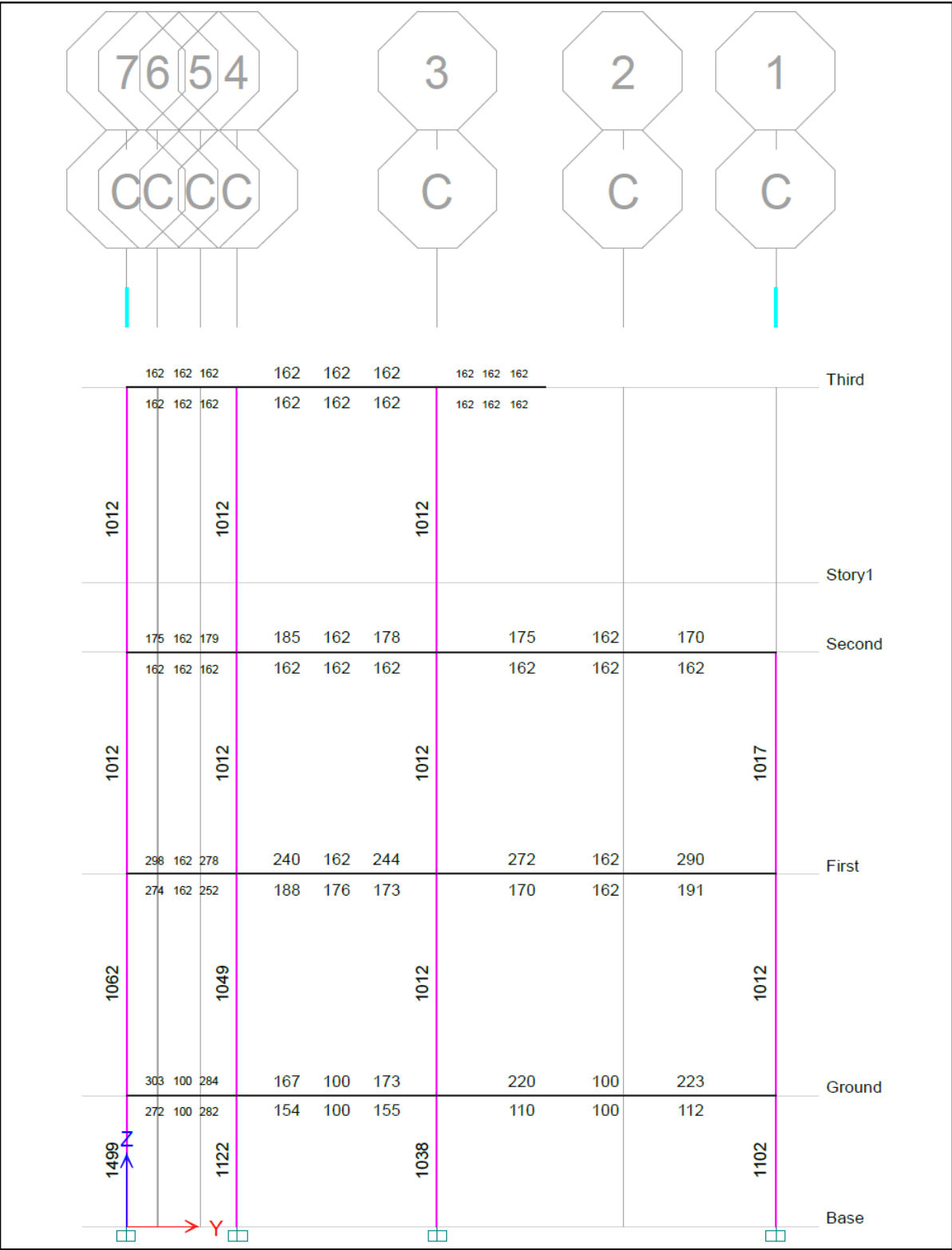


RunModel.EDB

Elevation View - A Longitudinal Reinforcing (IS 456:2000)

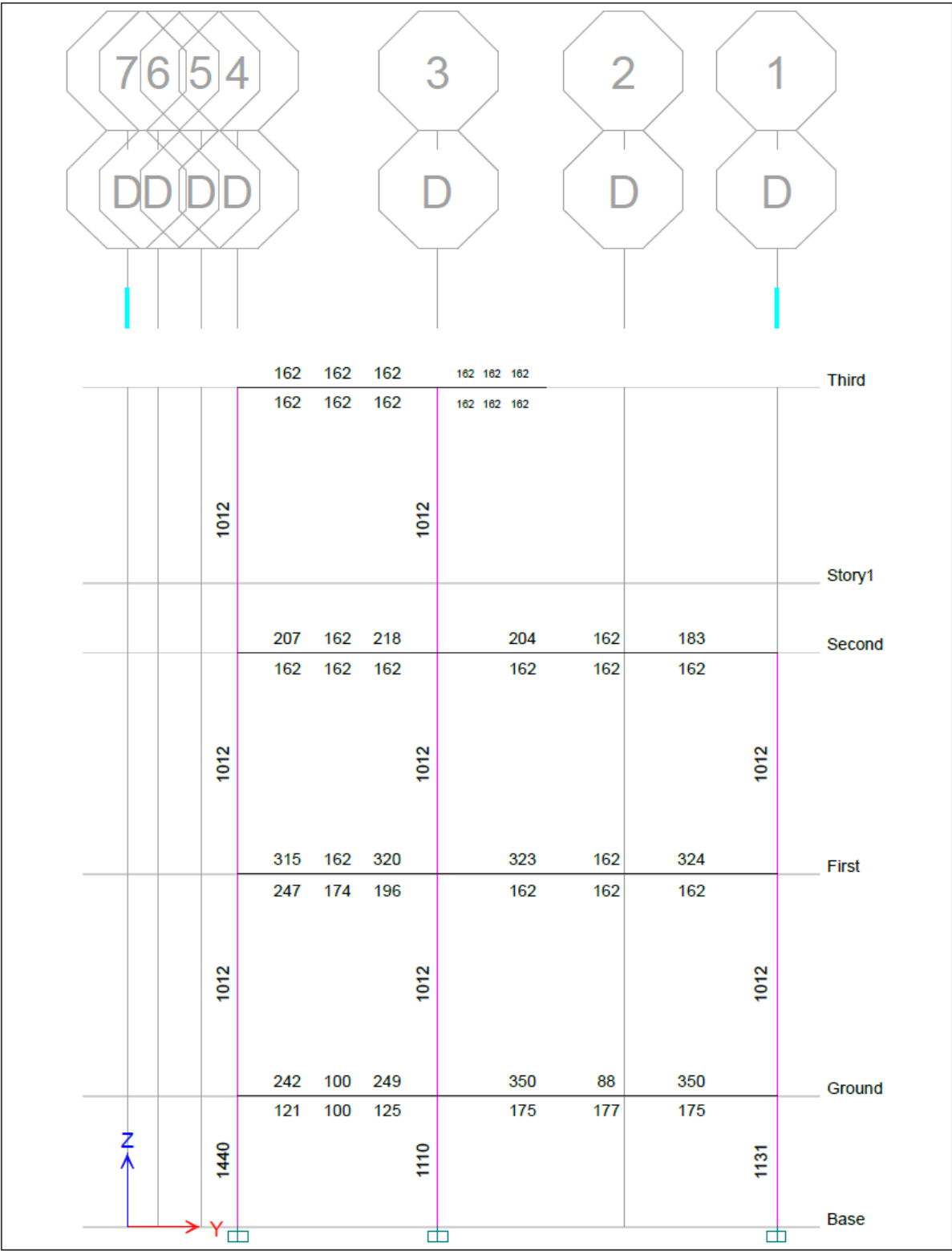
12/15/2021





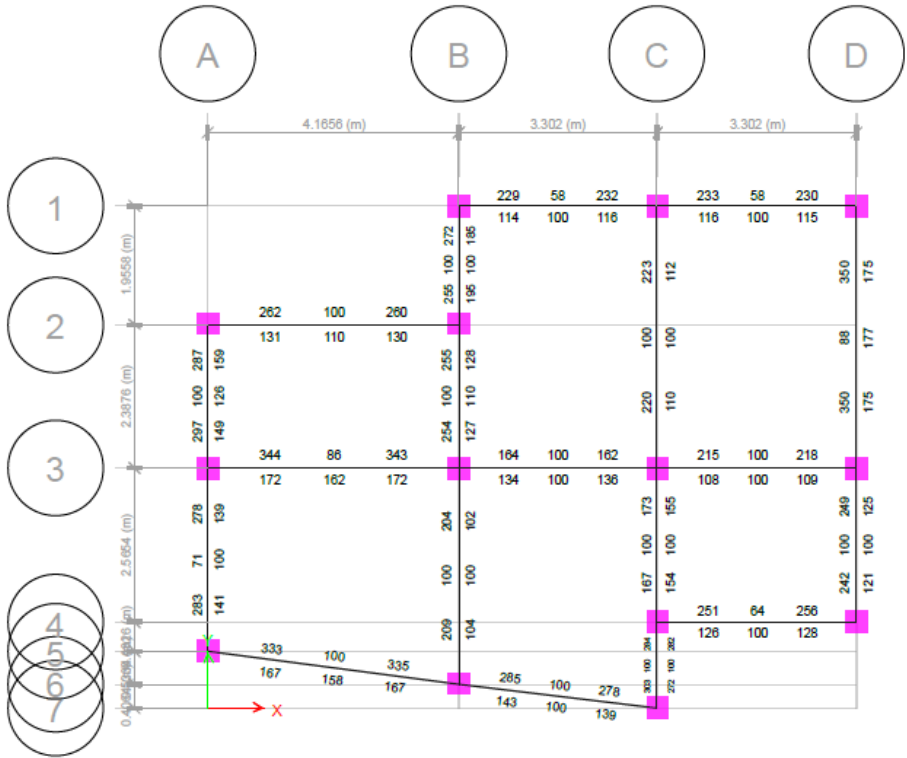
RunModel.EDB

Elevation View - C Longitudinal Reinforcing (IS 456:2000)

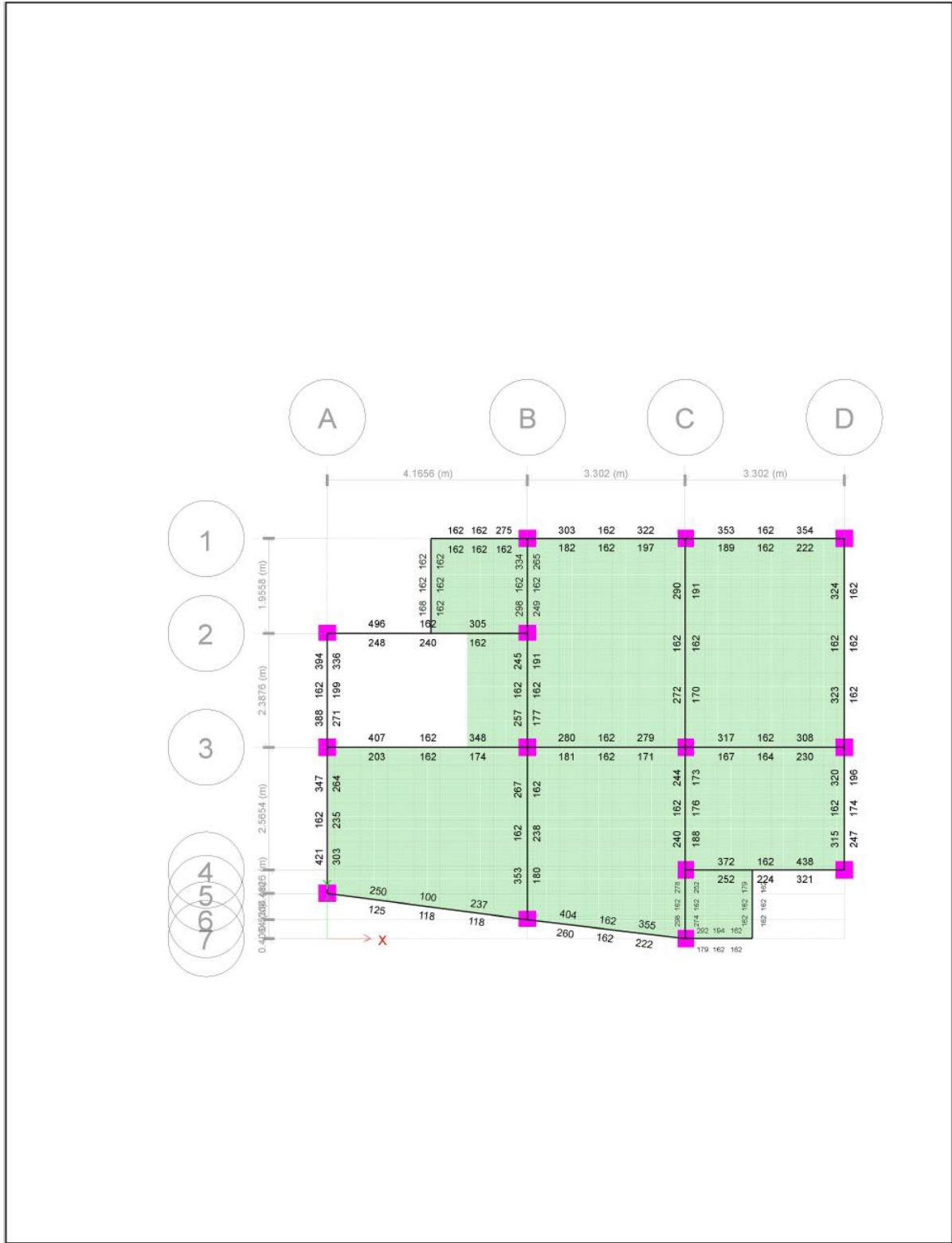


RunModel.EDB

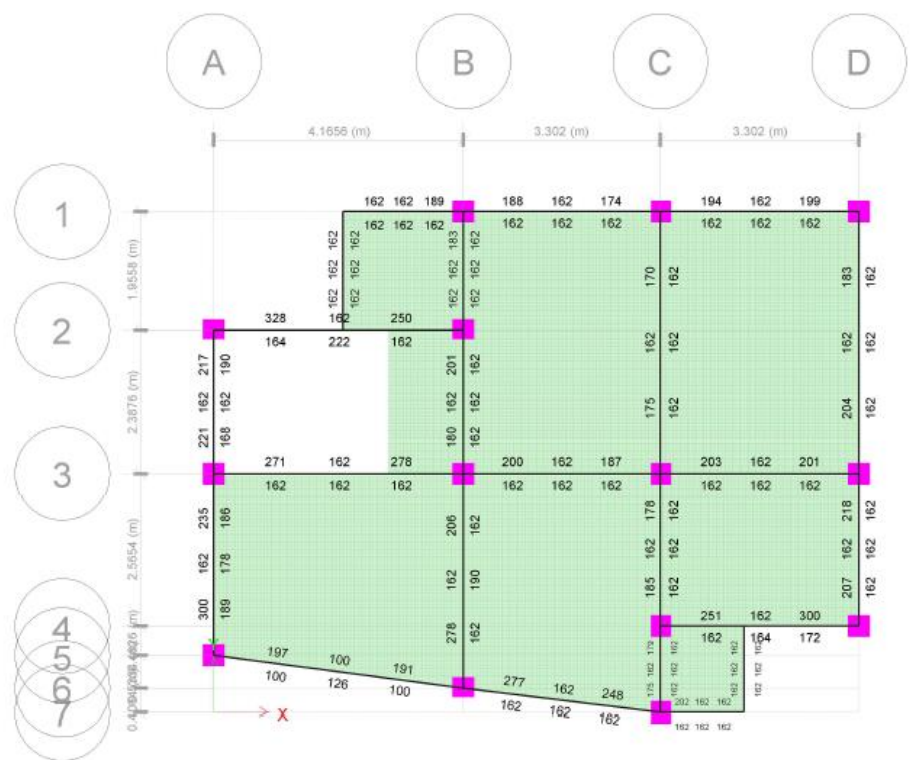
Elevation View - D Longitudinal Reinforcing (IS 456:2000)



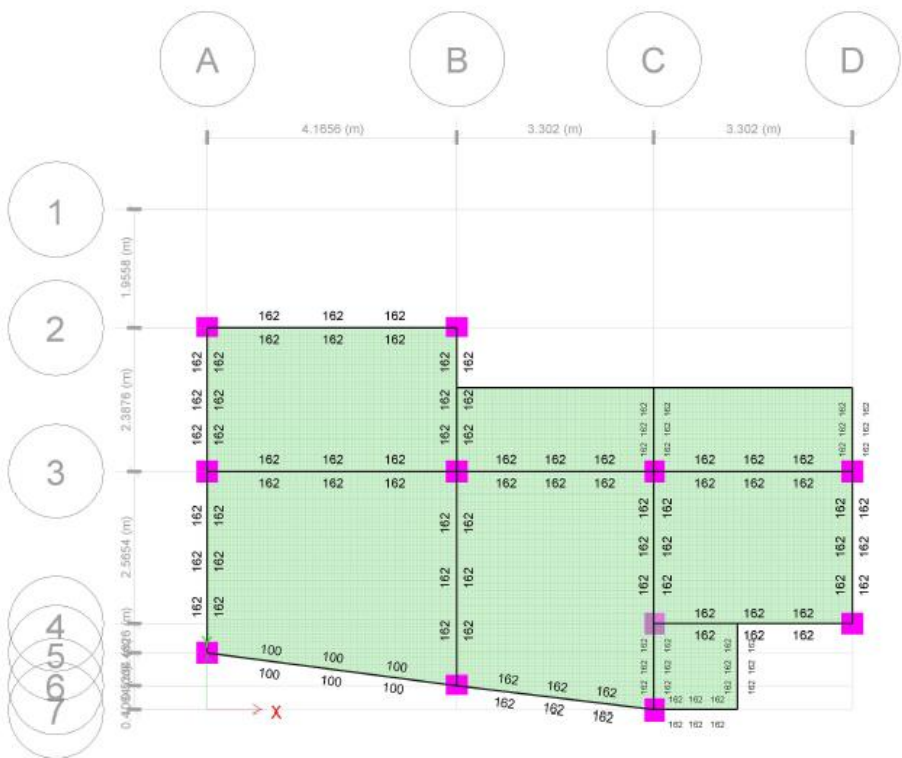
RunModel.EDBlAn View - Ground - Z = 1.6764 (m) Longitudinal Reinforcing (IS 456:2000)



RunModel.EDBPlan View - First - Z = 4.5212 (m) Longitudinal Reinforcing (IS 456:2000)



RunModel.EDB Plan View - Second - Z = 7.366 (m) Longitudinal Reinforcing (IS 456:2000)



RunModel.EDB Plan View - Third - Z = 10.7696 (m) Longitudinal Reinforcing (IS 456:2000)